

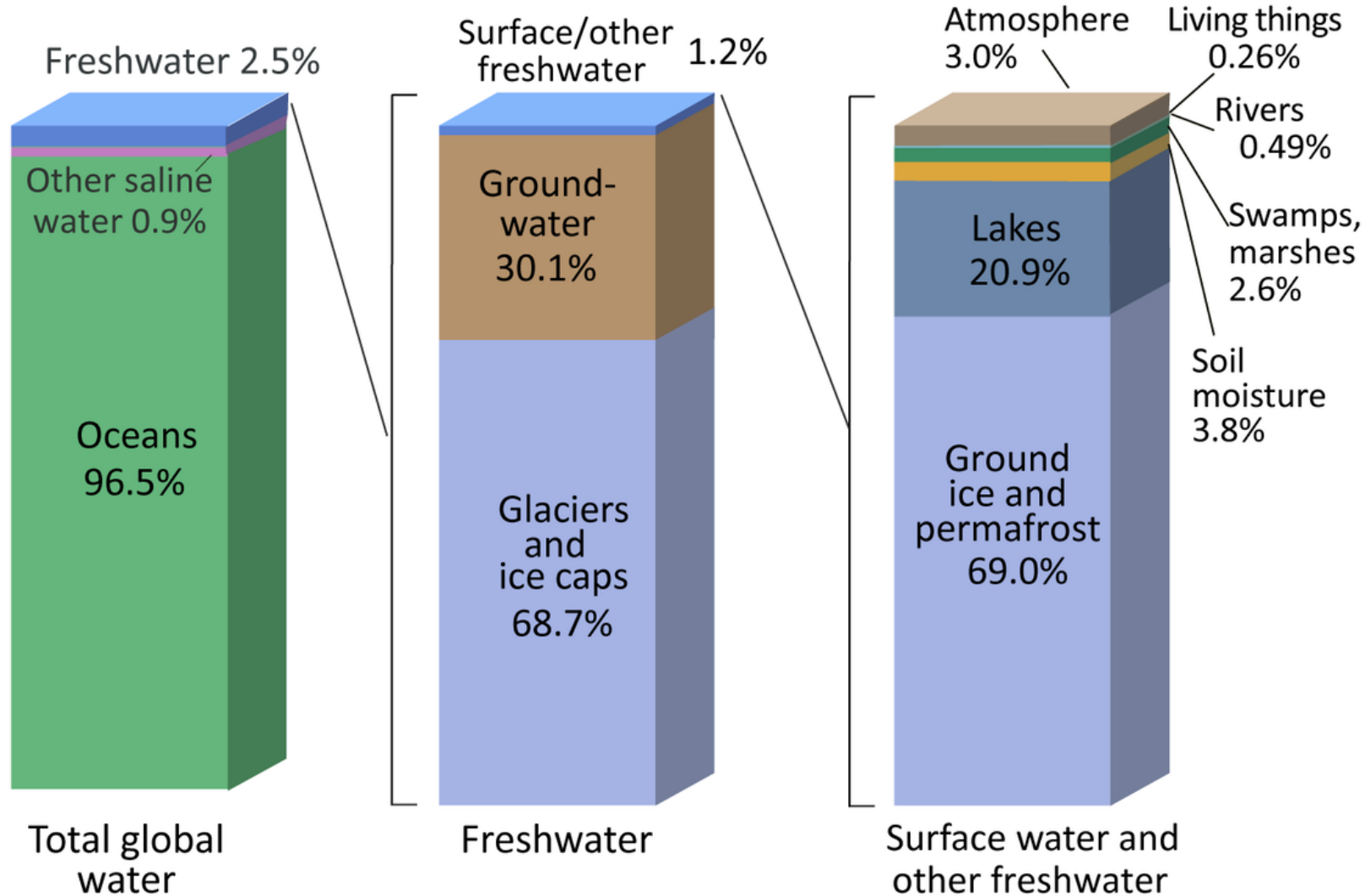
EES102: Introduction to Environmental Sciences

Water

Hydrological cycle, Water resource conservation,
Rainwater harvesting methods, Regulation of
Water Quality, Water Treatment and purification,
Water footprint of consumer products

by
Dr. Sunil A. Patil

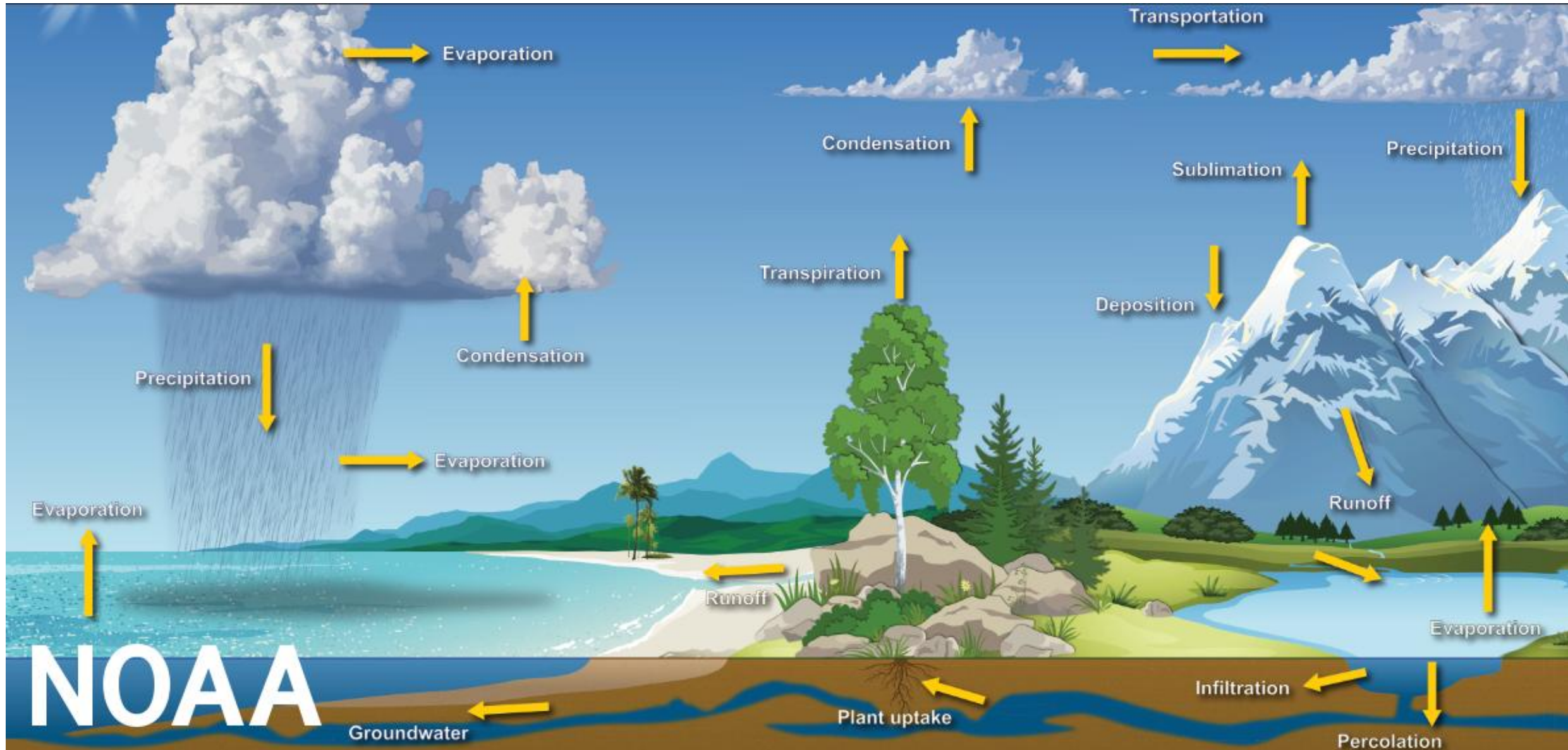
Where is Earth's Water?

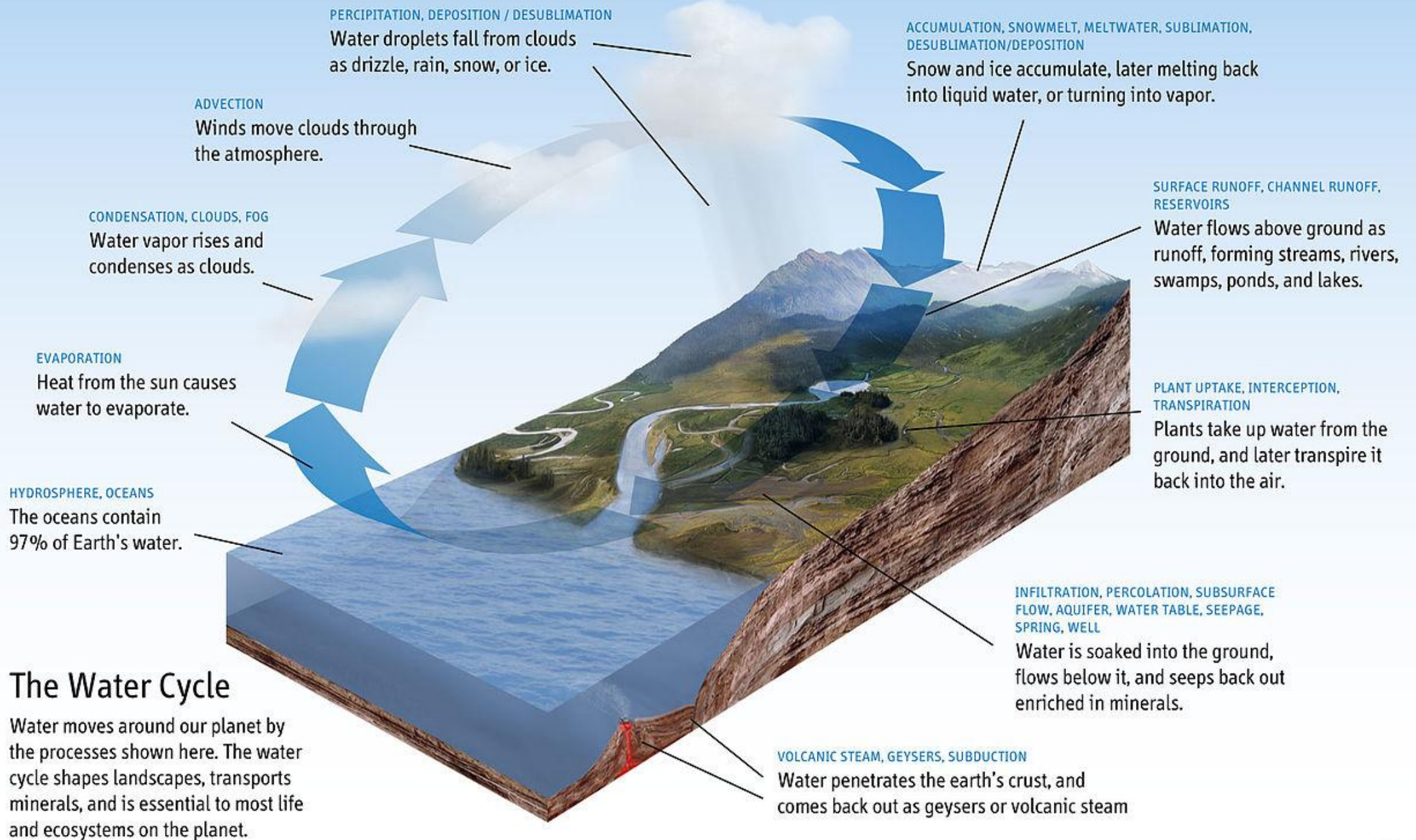


Source:
[https://www.usgs.gov/
media/images](https://www.usgs.gov/media/images)

Credit: U.S. Geological Survey, Water Science School. <https://www.usgs.gov/special-topic/water-science-school>
Data source: Igor Shiklomanov's chapter "World fresh water resources" in Peter H. Gleick (editor), 1993, Water in Crisis: A Guide to the World's Fresh Water Resources. (Numbers are rounded).

Water cycle – key processes





Water Reservoirs and Residence Time

- **Reservoirs** – locations where water exists at any point in the water cycle
- **Residence time** – time for which a water molecule remains in one reservoir

For steady systems,

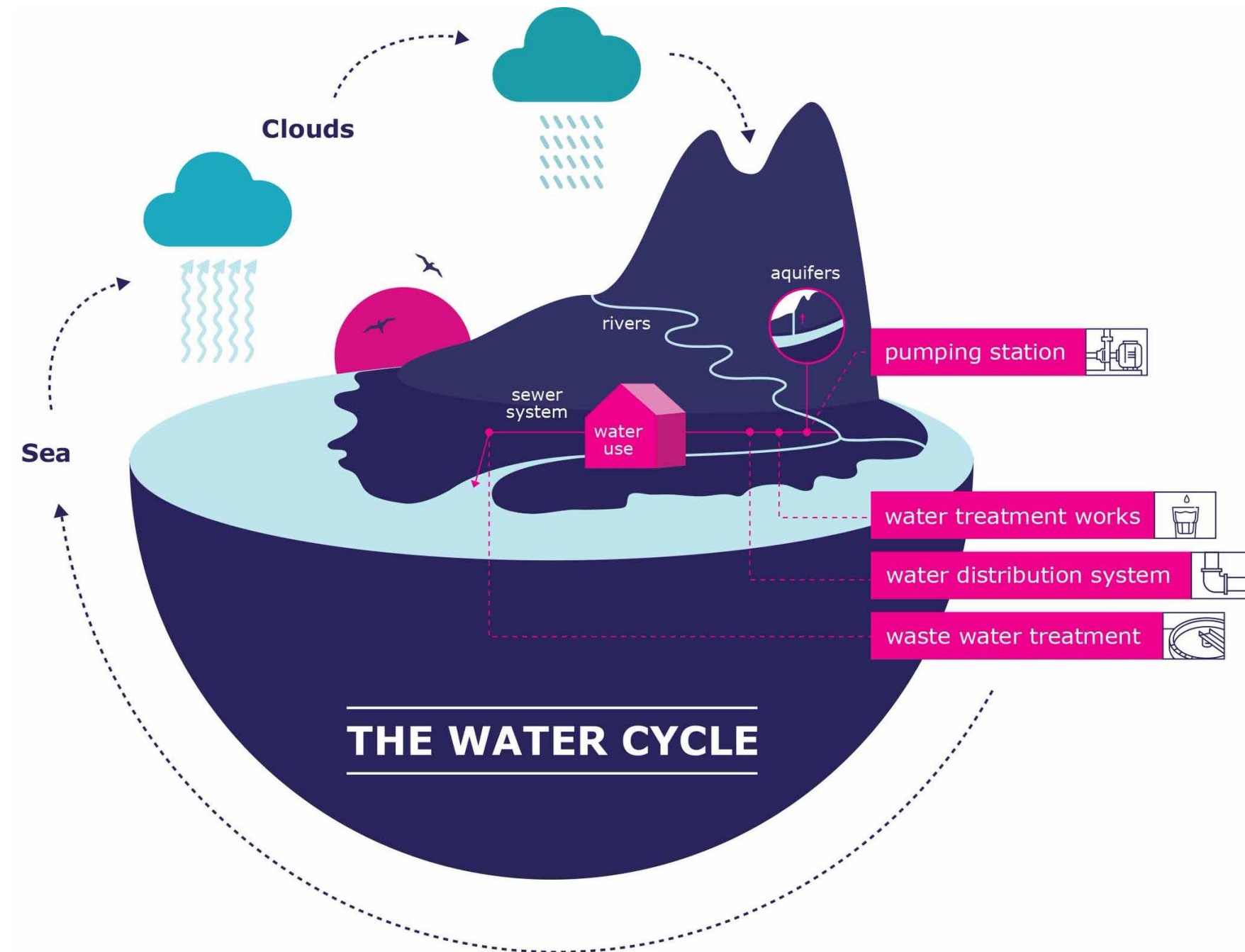
$$Tr = V/Q \text{ (I or O)}$$

[Tr: Residence time; V: Volume of the reservoir, Q (I or O): Mean inflow or outflow rate]

- The residence time of water in the atmosphere is the shortest of all — about 9 days.
- In the ocean it is about 3,200 years; of fossil water, i.e., ancient groundwater reservoirs, can be 1000s of years.

Reservoir	Average Residence Time
Glaciers	20 to 100 years
Seasonal Snow Cover	2 to 6 months
Soil Moisture	1 to 2 months
Groundwater: Shallow	100 to 200 years
Groundwater: Deep	10,000 years
Lakes	50 to 100 years
Rivers	2 to 6 months

<http://www.physicalgeography.net/fundamentals/8b.html>



Key human interventions in the water cycle

- Withdrawal, storage, extraction, deforestation, climate change, etc.
- Freshwater and wastewater management
- Includes water use for various activities, e.g., agriculture, industrial processes, human consumption

Water demand

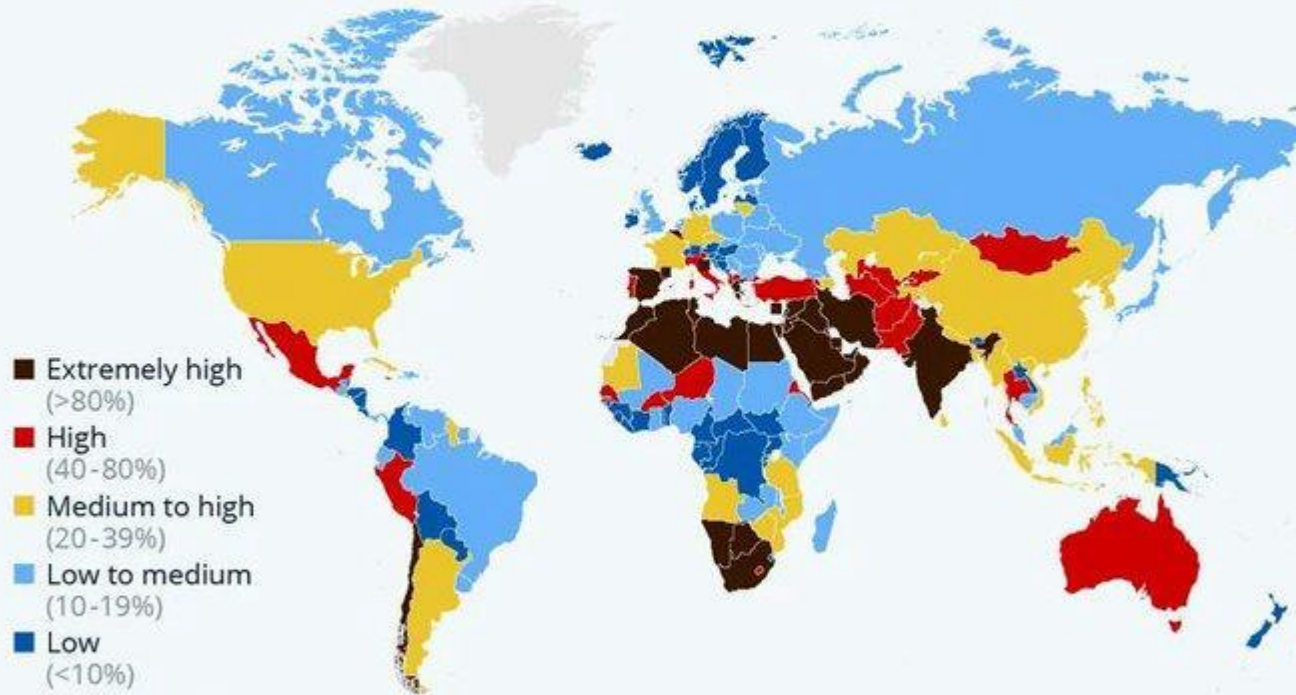
- Agriculture: 70%, Industrial: 20 % and Domestic: 10 %
- Global withdrawals **have tripled over the last 50 years** to meet the constant freshwater needs.
- Under an average economic growth scenario, the global freshwater requirements will grow **from 4,500 billion m³ to 6,900 billion m³ by year 2030** (Source: the 2030 Water Resource Group).
- The predicted figure for 2030 exceeds the current accessible and reliable supplies by 40 %.
- **A gap between available resource/supply and demand is evident.**

<https://2030wrg.org/>

Why effective water management is required?

Where Water Stress Will Be Highest by 2050

Projected ratio of human water demand to water availability (water stress level) in 2050*



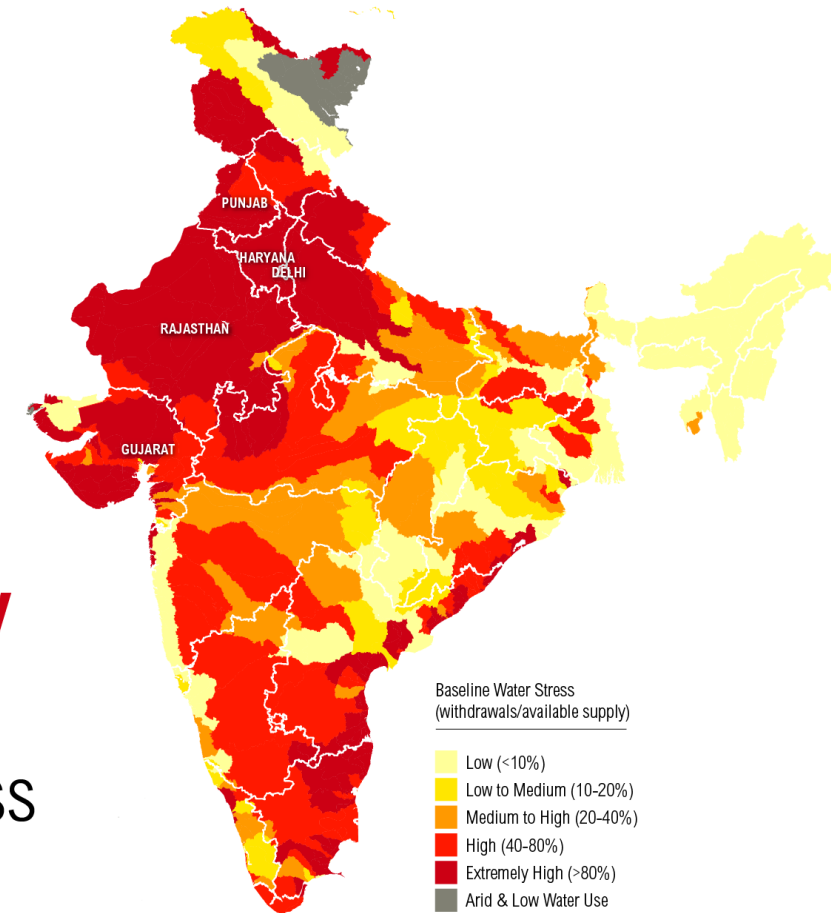
* According to "business as usual" scenario=middle-of-the-road future where temperatures increase by 2.8°C to 4.6°C by 2100

Source: World Resources Institute



statista

54%
of India
Faces
**High to
Extremely
High**
Water Stress



www.indiawatertool.in

WORLD RESOURCES INSTITUTE

Water conservation

- It involves efforts to waste less (fresh)water, optimize water use, and preserve water availability
- It is an important and integral part of long-term water resource planning/management
- Different recommended practices according to the sectors
 - agriculture irrigation, industries, etc.,
 - communities, households and policymakers/water utility authorities

Individuals, households, communities, etc.

- **At individual level – use as per requirement – avoid wastage – reuse as much as possible**
- Toilet systems to reduce water consumption (from 20 L to <1L per flush)
- Low flow shower heads (<5 L/min flow rate)
- Tap/faucet aerators (<6 L/min flow rate)
- WaterSense appliances: Water-efficient dishwasher (from 50 L to <10 L per cycle), washing machine (from 150 L to 50 L per cycle)
- Repairing leaks
- Native plants



ARE WE DOING ENOUGH TO SAVE WATER?



Avoid water leakage in faucet



Do not let fresh water run



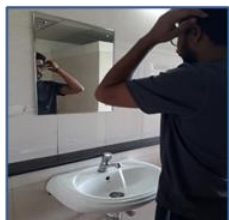
Run a full load in washing machine



Avoid overflow from the storage tanks



Use shower for < 2 min for bathing



Keep the tap off, if not needed



Ask for half a glass of water to avoid wastage



Avoid bathtub



Use bucket for car washing



- Check for toilet leakages
- Don't flush things down the toilet to dispose them
- Toilet top sink concept should be encouraged



Poster made by our lab for Jal Shakti Abhiyan 2019

TOP 5 WAYS TO SAVE WATER IN YOUR HOME



Take a shower of 5 minutes or less

Saves up to
70 Litres
per shower



Install a water-saving toilet flush system

Saves up to
700 Litres
each year



Turn off the tap while brushing your teeth and washing your hands

Saves up to
20 Litres
per day



Don't keep the tap running while washing clothes/utensils

Saves more than
50 Litres
per day



Close taps properly and fix leaking taps, pipes and toilets

Saves up to
75 Litres
each day

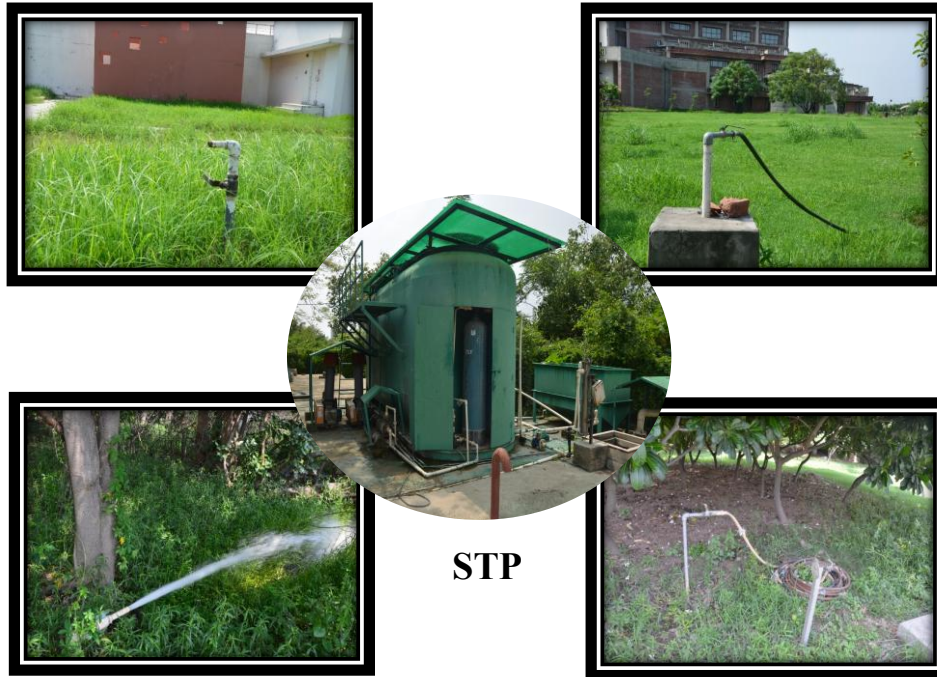
<https://www.alfaaav.com/blog/top-5-ways-to-save-water-in-your-home/>

At community level

- System Improvements — Keep a tight system, look at alternative sources
- Water-efficient infrastructure upgrades
- Policies and Programs to Encourage Efficient Water Use – Water meters, reuse/recycle, watershed management, audits/monitoring, community gardening, incentives, etc.
- Rainwater harvesting
- Public awareness campaigns

Water Conservation Practices at our Institute

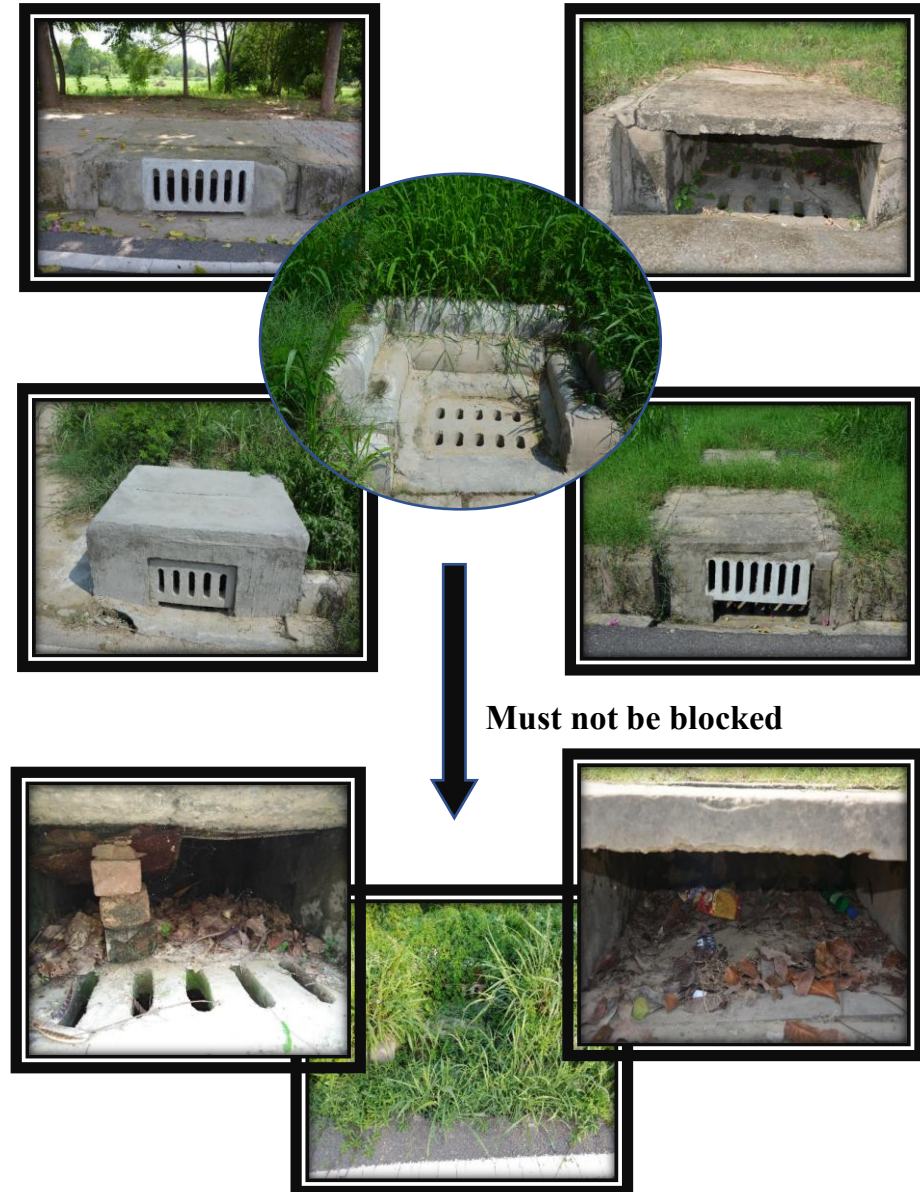
Treated Water Reuse



Ground Water Recharge systems



Rain Water collection points

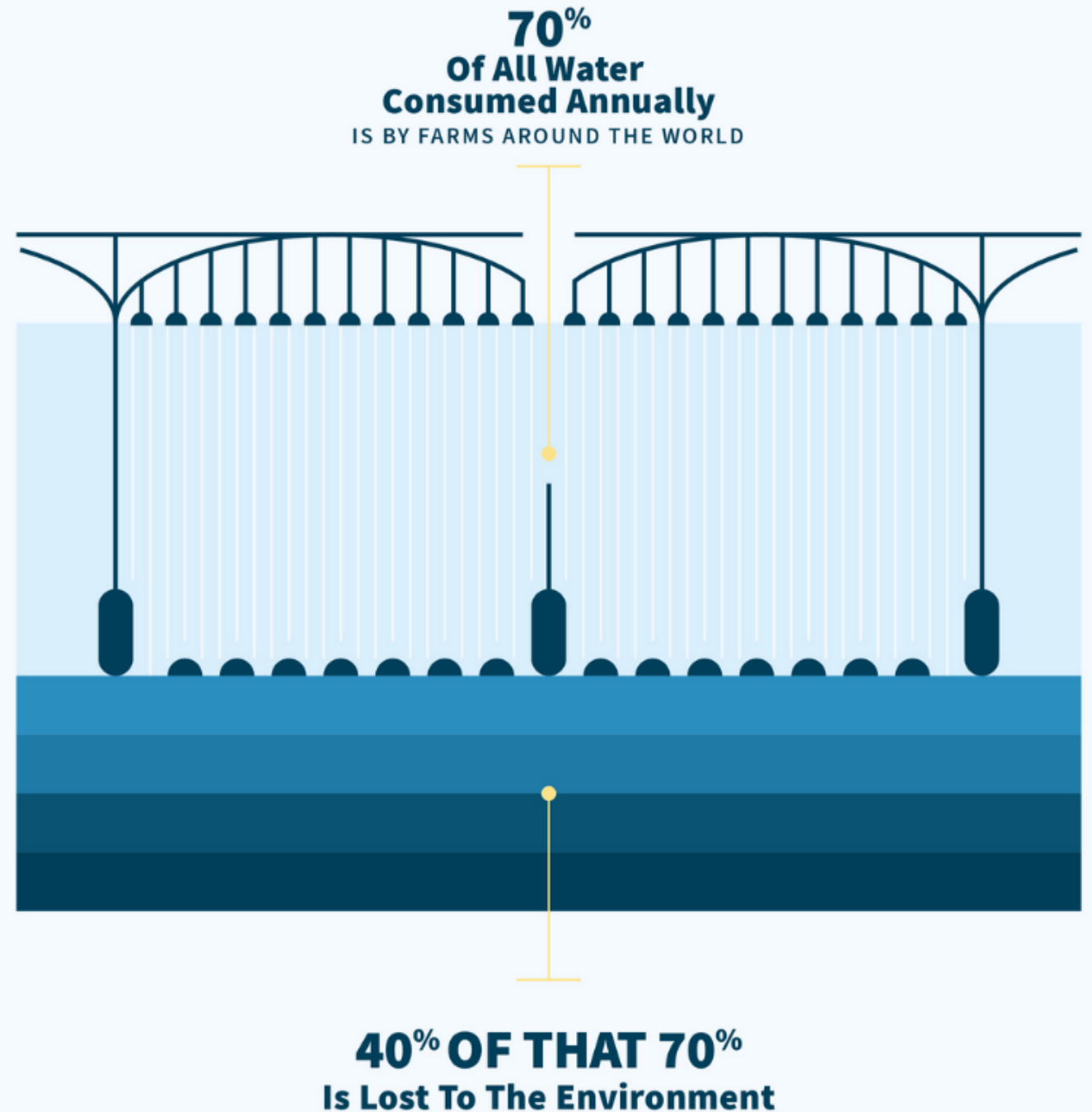


Collage by our lab for
Jal Shakti Abhiyan 2019

Agriculture

- Irrigation of farm lands
- Cattle farming (e.g., dairy)

<https://htt.io/water-usage-in-the-agricultural-industry/>



Micro-irrigation practices

- 60% less water consumption as compared to the traditional irrigation system
- Less fertilizers use (e.g., fertigation)
- Improved productivity and quality
- Reduced weed congestion



Drip irrigation



Sprinkler

<https://krishijagran.com/>



Rain gun

<https://pangonionlinemarketing.com/>

Industries

- Thermal power plants, food and beverages, chemicals, paper, textile, cement, iron and steel, etc.
- Changing behavior
- Routine checks and maintenance; process optimization
- Water audits
- Modifying and/or replacing existing equipment with water saving equipment to reduce overall water consumption
- Increasing internal water reuse

Water utility authorities

- Meter all water (metered customers use less water - up to 45 % - than the unmetered customers)
- Water accounting and loss control
- Pricing water (real cost: current + future supply costs)
- Information and Education

Credit: Jeffrey Beall
The Xeriscape Demonstration Garden, located at the headquarters of Denver Water in Denver, Colorado



The United Nations World Water Development Report 2021

VALUING WATER



Rainwater harvesting (RWH)

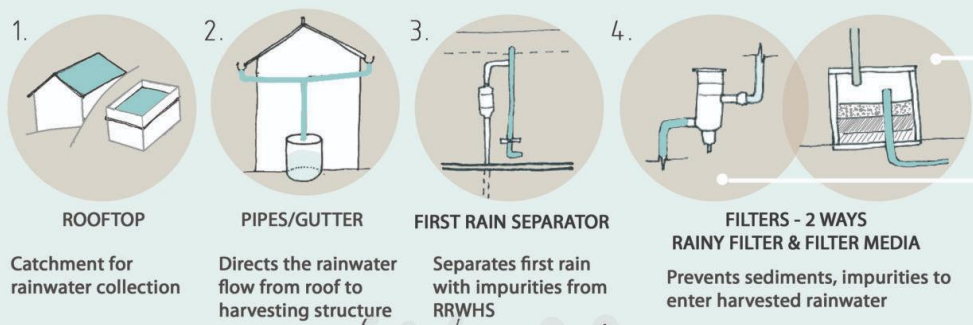
- Collection and efficient storage of rainwater from different basement areas into natural reservoirs or artificial tanks, or the infiltration of surface water into subsurface aquifers (recharging groundwater) before it is lost as surface runoff
- Two basic ways: Rooftop (in tanks) and surface runoff (in check dams and underground tanks) harvesting
- From simple techniques such as using utensils to intricate designs such as check dams (also applicable to water conservation)

Major components of RWH system

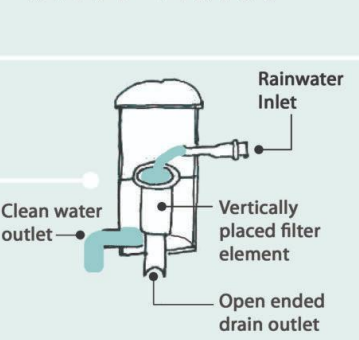
- Catchment area
- Conveyance system
- First flush device
- Filtration system (e.g., sand, charcoal, etc.)
- Storage tank
- Distribution system



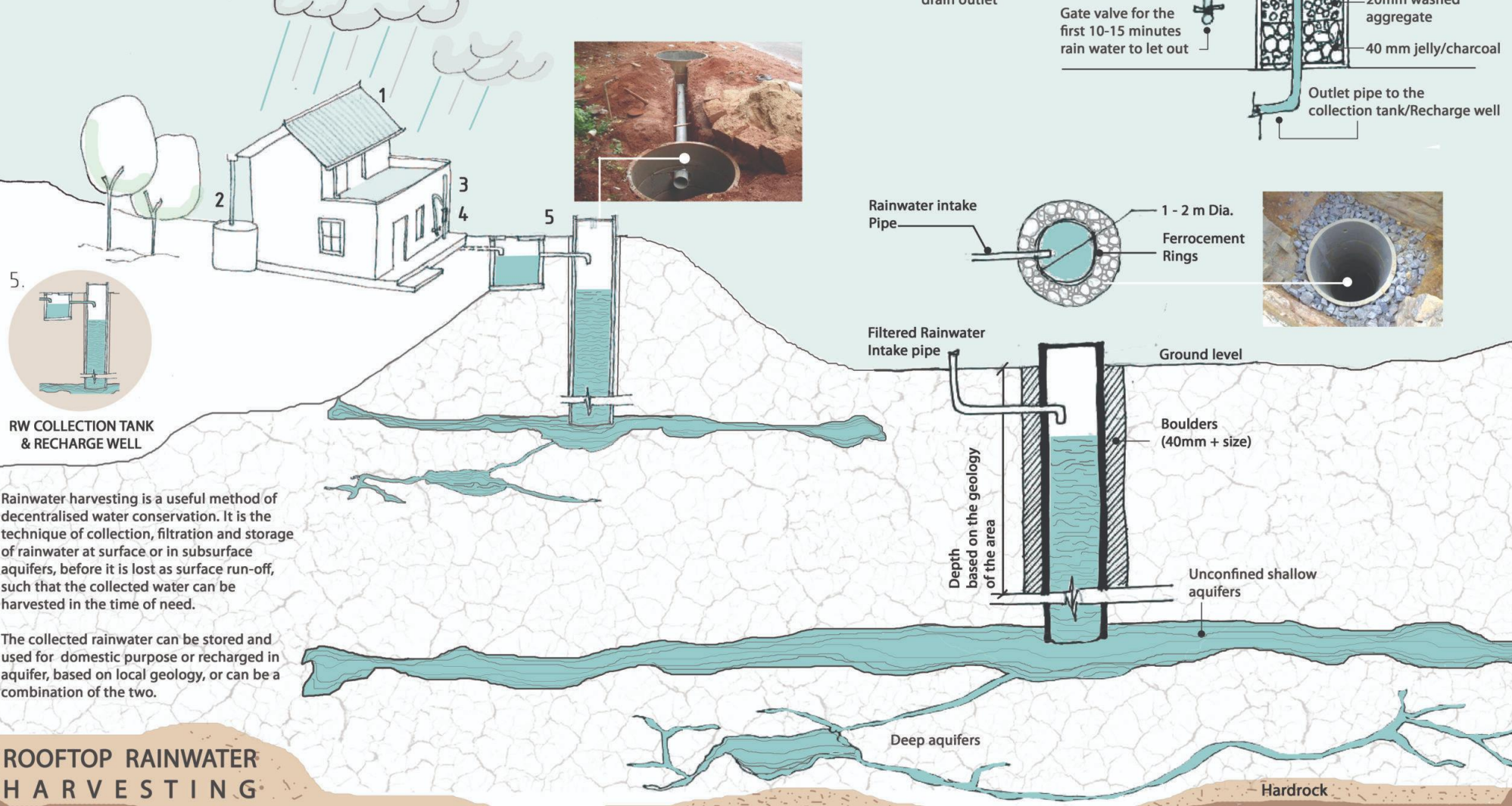
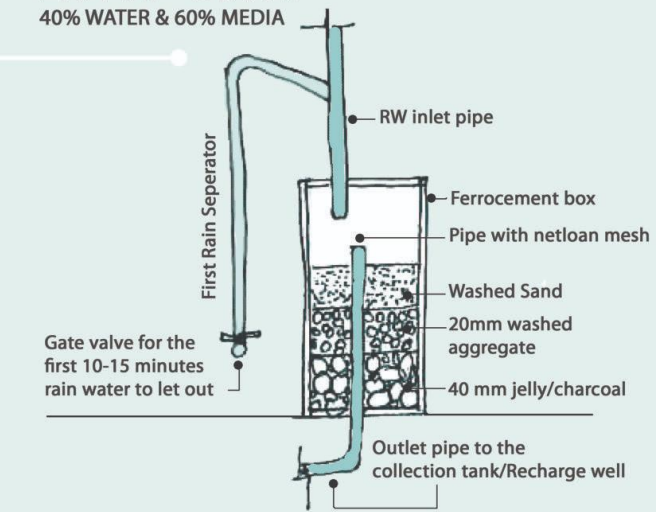
COMPONENTS OF RWH



RAINY FILTER



FILTER MEDIA
40% WATER & 60% MEDIA



Rainwater harvesting is a useful method of decentralised water conservation. It is the technique of collection, filtration and storage of rainwater at surface or in subsurface aquifers, before it is lost as surface run-off, such that the collected water can be harvested in the time of need.

The collected rainwater can be stored and used for domestic purpose or recharged in aquifer, based on local geology, or can be a combination of the two.

ROOFTOP RAINWATER HARVESTING

Rainwater harvesting (RWH) system components (E.g., rooftop)

Source: WIN foundation



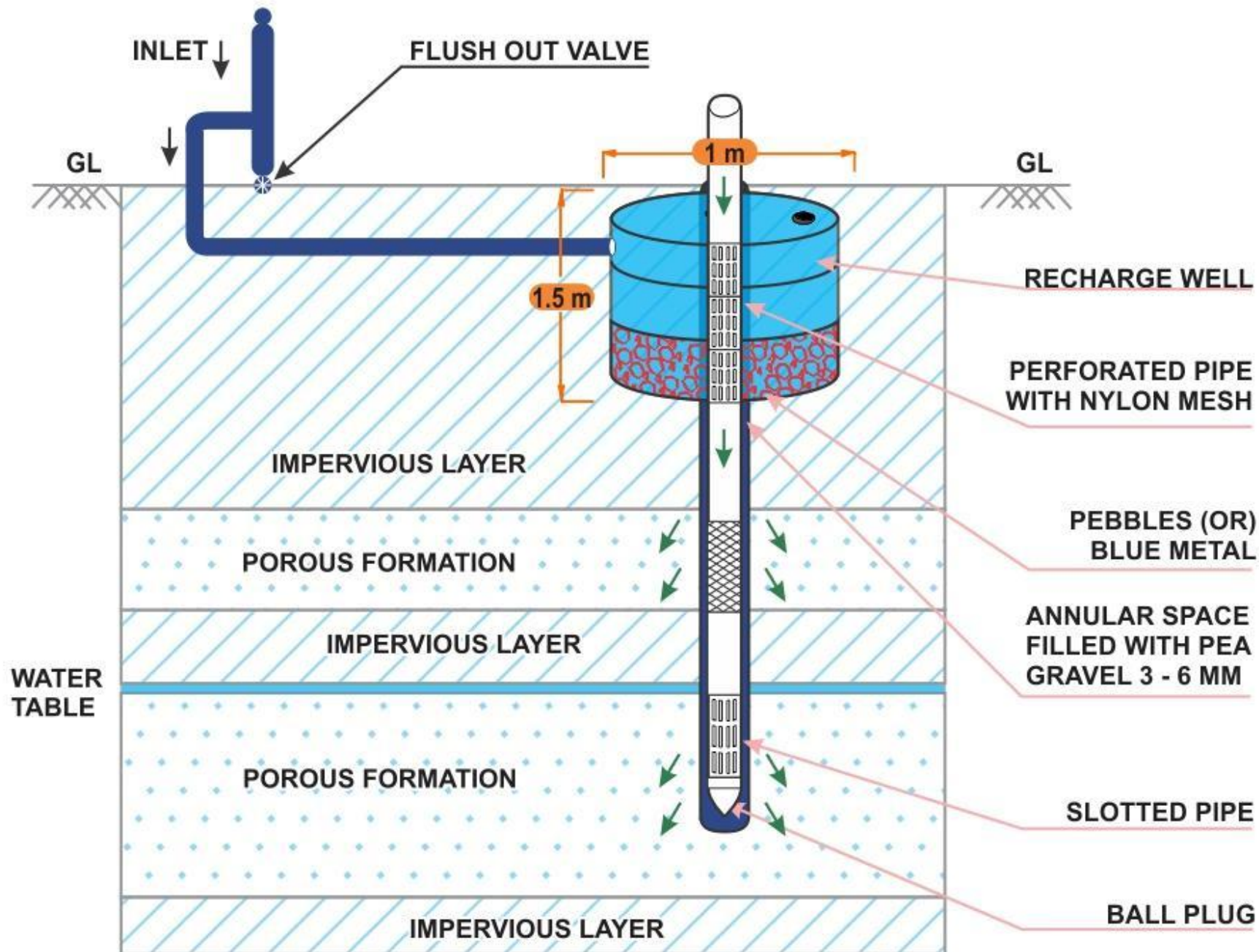
Recharge
pit

<http://www.rainwaterharvesting.org/Rural/Rural.htm> for rainwater harvesting methods in India

- **Total rainwater collection (liters) =**
Mean annual rainfall (in mm) X roof/catchment area (m²)
X runoff factor/coefficient

Runoff coefficient

- Range: 0 to 1.0 (pervious to impervious surfaces)
- Generally, about 0.8-0.9 for concrete roofs and built-up areas



Groundwater recharge system

<https://www.twadboard.tn.gov.in/>

Rainwater harvesting @IISER Mohali (#18 units)

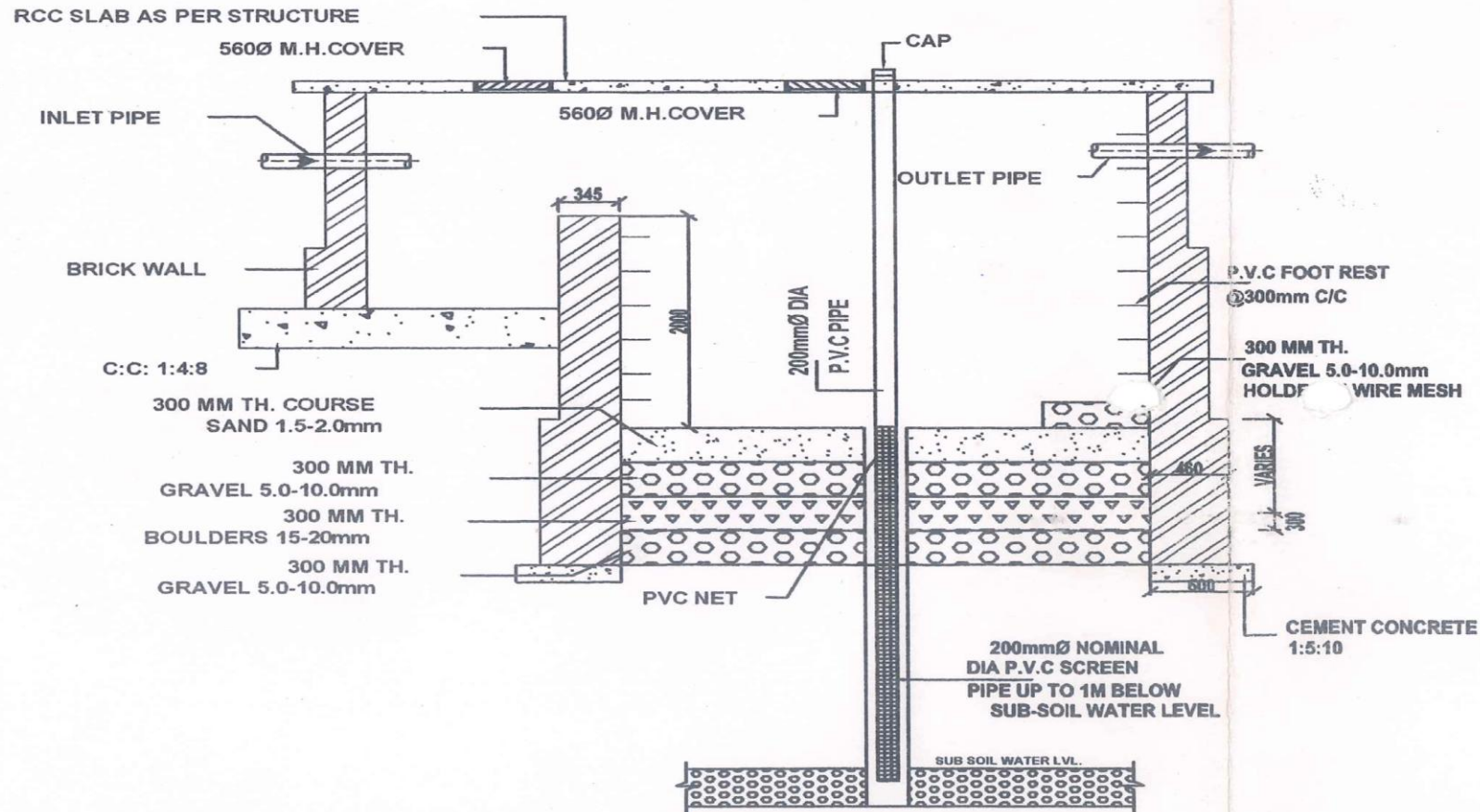
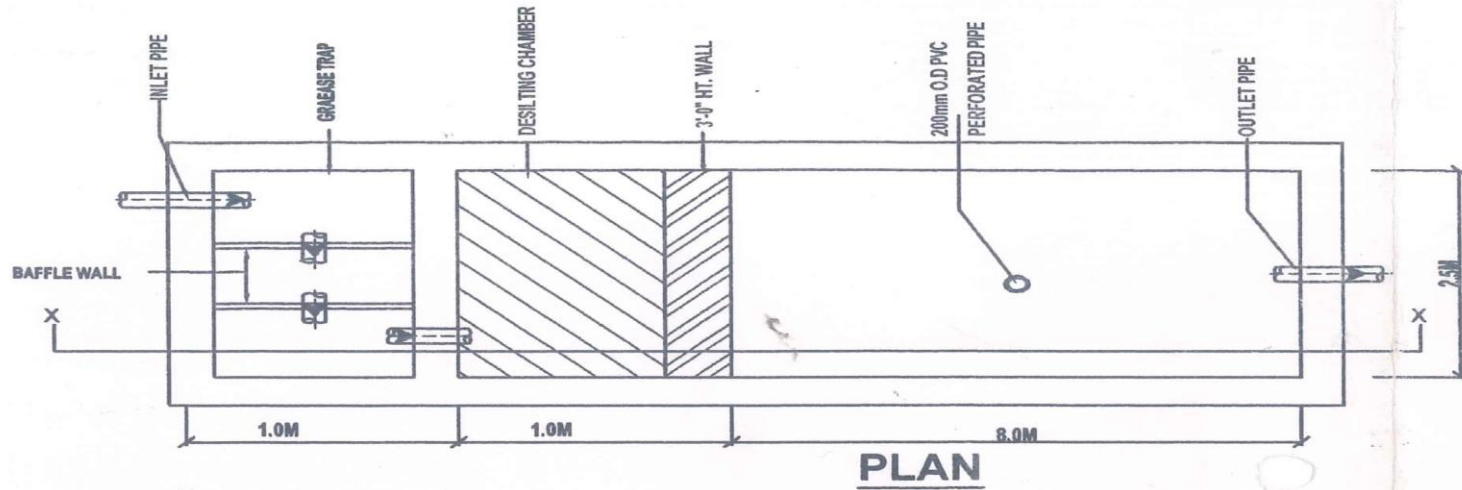




Image: EPS



(Image courtesy: Hindi Water Portal)



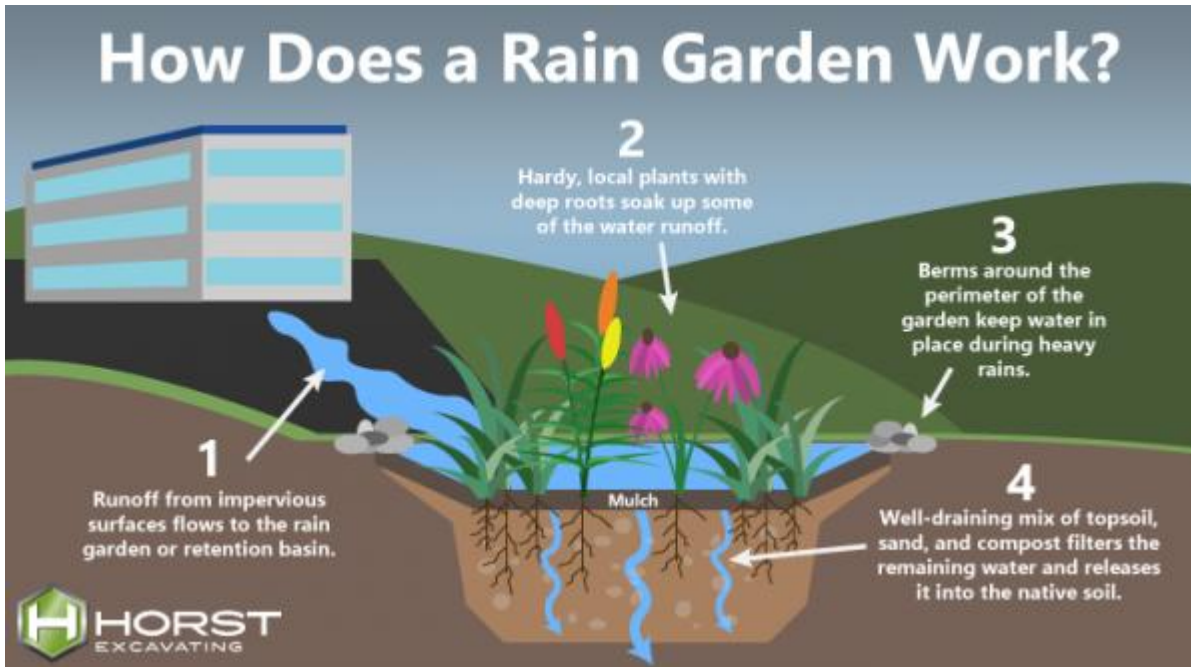
Image source: The Hindu



ANI

Rain gardens and Green roofs

Components of green roofs



DOI: 10.13140/RG.2.2.10839.32163





WATER OPTIONS

World Water Day 2021: Ten Positive Water Stories from India

Search ...

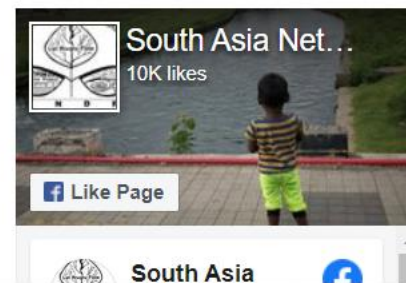
Archives

Select Month ▾

Categories

Select Category ▾

SANDRP Facebook Page



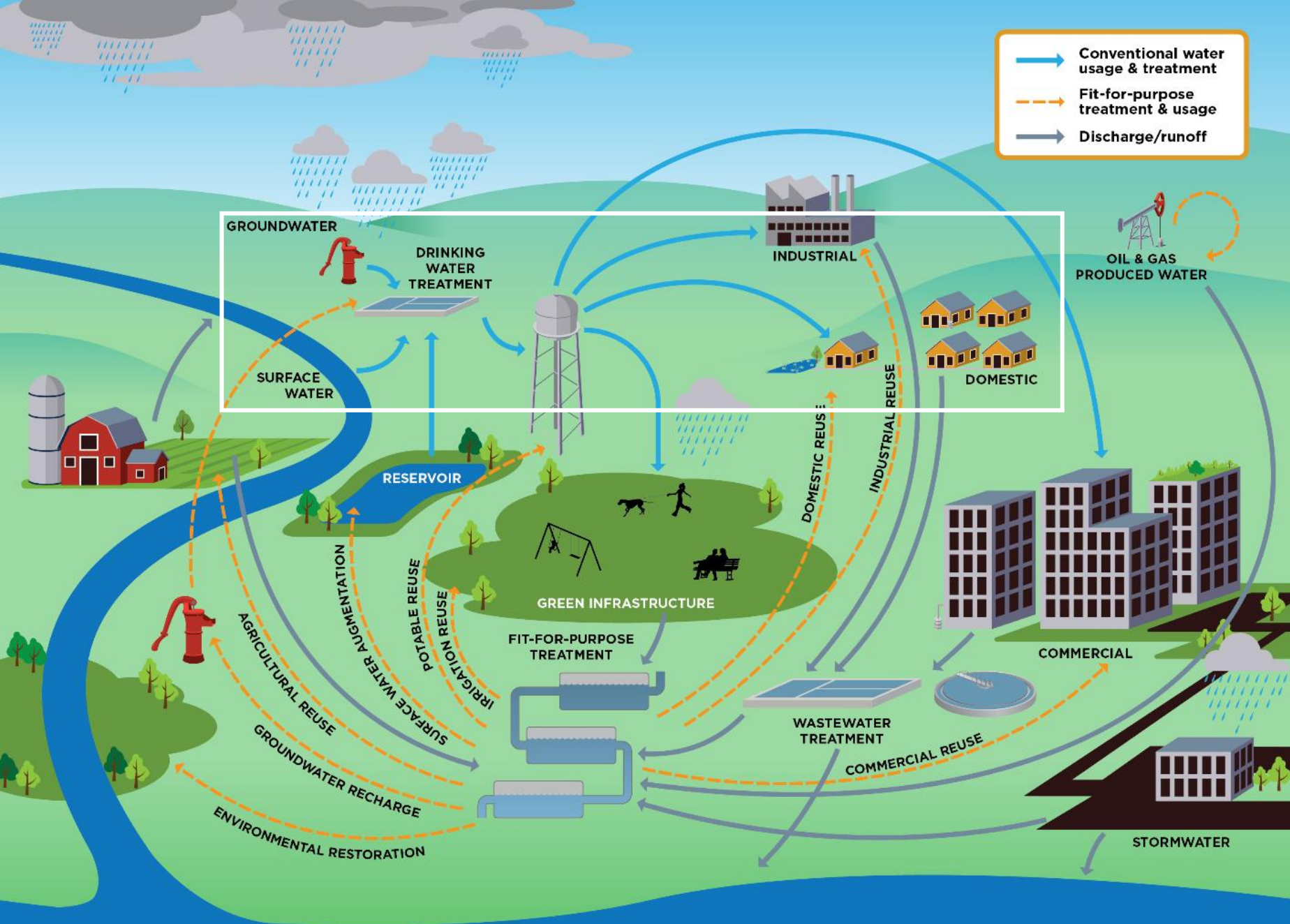
Additional reports:

<https://thebetterindia.com/416170/india-water-conservation-efforts-kashmir-kanyakumari-map/>

<https://waterconservation.rtofliving.org/impact-river-rejuvenation.php>

<https://www.indiawaterportal.org/drinking-water/women-leading-charge-water-transformation-and-conservation>

<https://sandrp.in/2021/03/21/world-water-day-2021-ten-positive-water-stories-from-india/>



Examples of water sources and use applications

<https://www.epa.gov/>

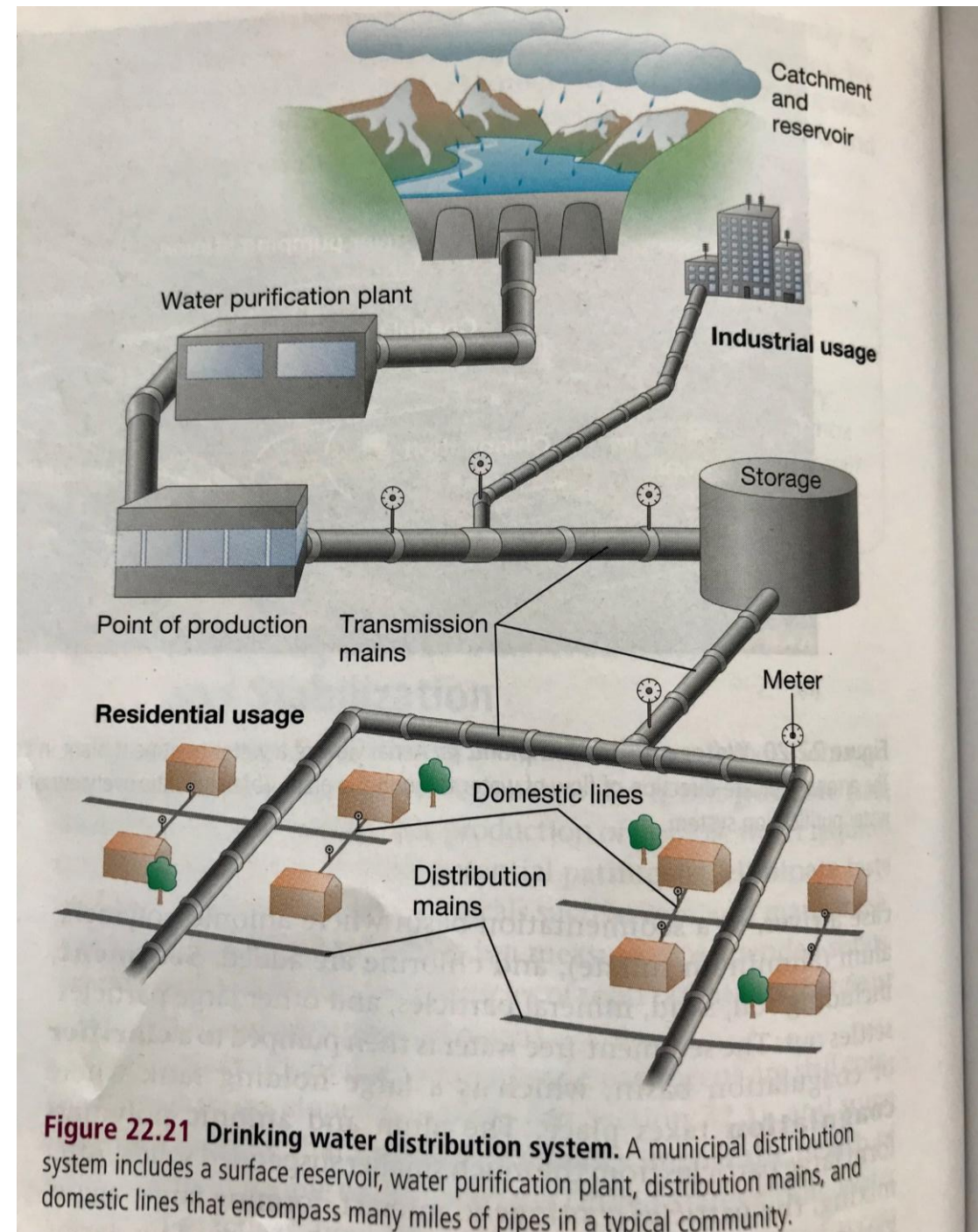
Potable/Drinking water

- For water to be regarded as potable, i.e. of a quality fit and safe for drinking, it:
 - should not have an unpleasant odor, taste and colour (aesthetic aspects)
 - must not contain any harmful chemicals
 - must be free from microbial pathogens
- Drinking water should not represent any significant risk (including any sensitivities) to human health over a lifetime of consumption.

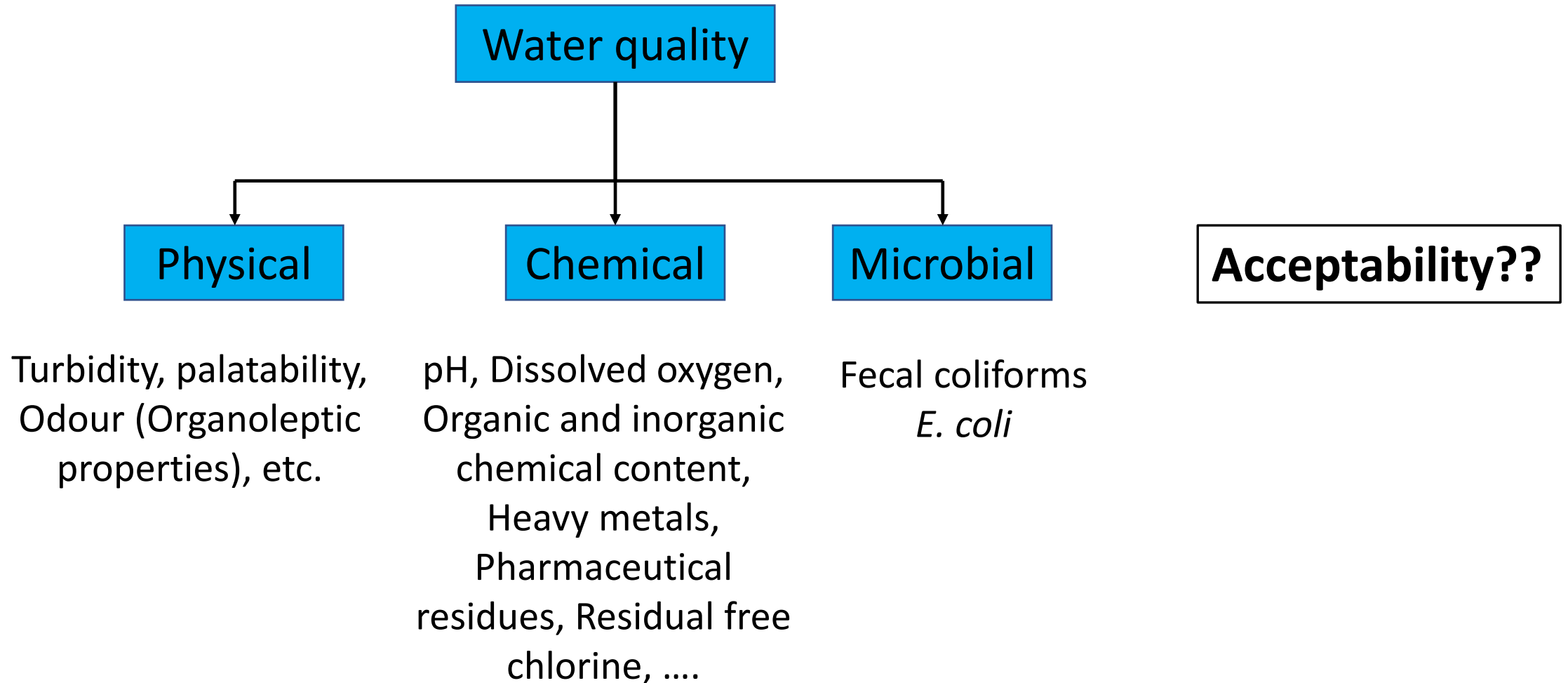
Most common water sources obtained from municipalities/gated communities have:

- been treated or purified
- high chemical and microbiological standards
- been tested on a pre-determined schedule

Brock Biology of
Microorganisms, 15th edition



Drinking water quality assessment



WATER QUALITY PARAMETERS AND BIS STANDARDS FOR VARIOUS CHEMICAL AND BIOLOGICAL CONSTITUENTS

Parameters	Drinking water IS 10500 : 2012	
	Permissible Limit	Maximum Limit
Odor	Agreeable	Agreeable
Taste	Agreeable	Agreeable
pH	6.5 to 8.5	No relaxation
TDS (mg/l)	500	2000
Hardness (as CaCO ₃) (mg/l)	200	600
Alkalinity (as CaCO ₃) (mg/l)	200	600
Nitrate (mg/l)	45	No relaxation
Sulfate (mg/l)	200	400
Fluoride (mg/l)	1	1.5
Chloride (mg/l)	250	1000
Turbidity (NTU)	1	5
Arsenic (mg/l)	0.01	No relaxation
Copper (mg/l)	0.05	1.5
Cadmium (mg/l)	0.003	No relaxation
Chromium (mg/l)	0.05	No relaxation
Lead (mg/l)	0.01	No relaxation
Iron (mg/l)	1.0	No relaxation
Zinc (mg/l)	5	15
Fecal Coliform (cfu)	0	0
E. Coli (cfu)	0	0

Test Results

S.No.	Parameter	Requirement (Desirable Limit) As per IS 10500 : 2012	Results	Units	Test Method
1	Color	Max 5	BDL(MDL:1)	Hazen	IS 3025:2021 (Part 4)
2	Odour	Agreeable	Agreeable		IS 3025: 2018 (Part 5)
3	pH	6.5-8.5	7.75		IS 3025:2022 (Part 11)
4	Taste	Agreeable	Agreeable		IS 3025: 2023 (Part 8) (RA 2023)
5	Turbidity	Max 1	BDL(MDL 0.5)	NTU	IS 3025 :2023 (Part 10) (RA 2023)
6	Total Dissolved Solids (TDS)	Max 500	73	mg/l	IS 3025 : 2023 (Part 16)
7	Chloride (as Cl)	Max 250	5.0	mg/l	IS 3025 (Part 32): 1998
8	Residual free chlorine	Min 0.2	BDL (MDL 0.1)	mg/l	IS 3025 (Part 26)
9	Iron (as Fe)	Max 1	BDL (MDL 0.01)	mg/l	APHA 3125 B By ICP-MS
10	Nitrate (as NO3)	Max 45	1.67	mg/l	IS 3025 (Part 34) : 2023
11	Selenium (as Se)	Max 0.01	BDL (MDL 0.005)	mg/l	APHA 3125 B (By ICP-MS)
12	Alkalinity Total (as CaCO3)	Max 200	28	mg/l	IS 3025 (Part 23): 2023
13	Hardness Total (as CaCO3)	Max 200	12	mg/l	IS 3025 (Part 21): 2009
14	Zinc (as Zn)	Max 5	BDL (MDL 0.005)	mg/l	APHA 3125 B (By ICP-MS)
15	Cadmium (as Cd)	Max 0.003	BDL (MDL 0.001)	mg/l	APHA 3125 B (By ICP-MS)
16	Lead (as Pb)	Max 0.01	BDL (MDL 0.005)	mg/l	APHA 3125 B (By ICP-MS)
17	Mercury (as Hg)	Max 0.001	BDL (MDL 0.0005)	mg/l	APHA 3125 B (By ICP-MS)
18	Molybdenum (as Mo)	Max 0.07	BDL (MDL 0.005)	mg/l	APHA 3125 B (By ICP-MS)
19	Nickel (as Ni)	Max 0.02	BDL (MDL 0.005)	mg/l	APHA 3125 B (By ICP-MS)
20	Arsenic (As)	Max 0.01	BDL (MDL 0.005)	mg/l	APHA 3125 B (By ICP-MS)
21	Chromium (as Cr)	Max 0.05	BDL (MDL 0.005)	mg/l	APHA 3125 B (By ICP-MS)
22	E.coli	Absent per 100ml	Absent	per 100ml	IS 15185: 2016
23	Coliform	Absent per 100ml	Absent	per 100ml	IS 15185: 2016

Representative water quality test report (AB-1)

BDL: below detection limit
MDL: method detection limit

Sources of drinking water

- Surface water (rivers, lakes, canals, ponds/reservoirs....)
- Groundwater
- Rainwater (harvested)
- Desalination of seawater
- Atmospheric water generation
- The desirable approach is to use water from the best available and safest source.

Prevention of drinking water contamination

- Source water protection
- **Reliable water treatment**
- Protection of the water distribution network

Impurities in raw water

- Can be grouped into three categories:

Physical: materials that do not dissolve in water and make the water appear "dirty".

Chemical: substances dissolved in the water from both natural and man-made processes.

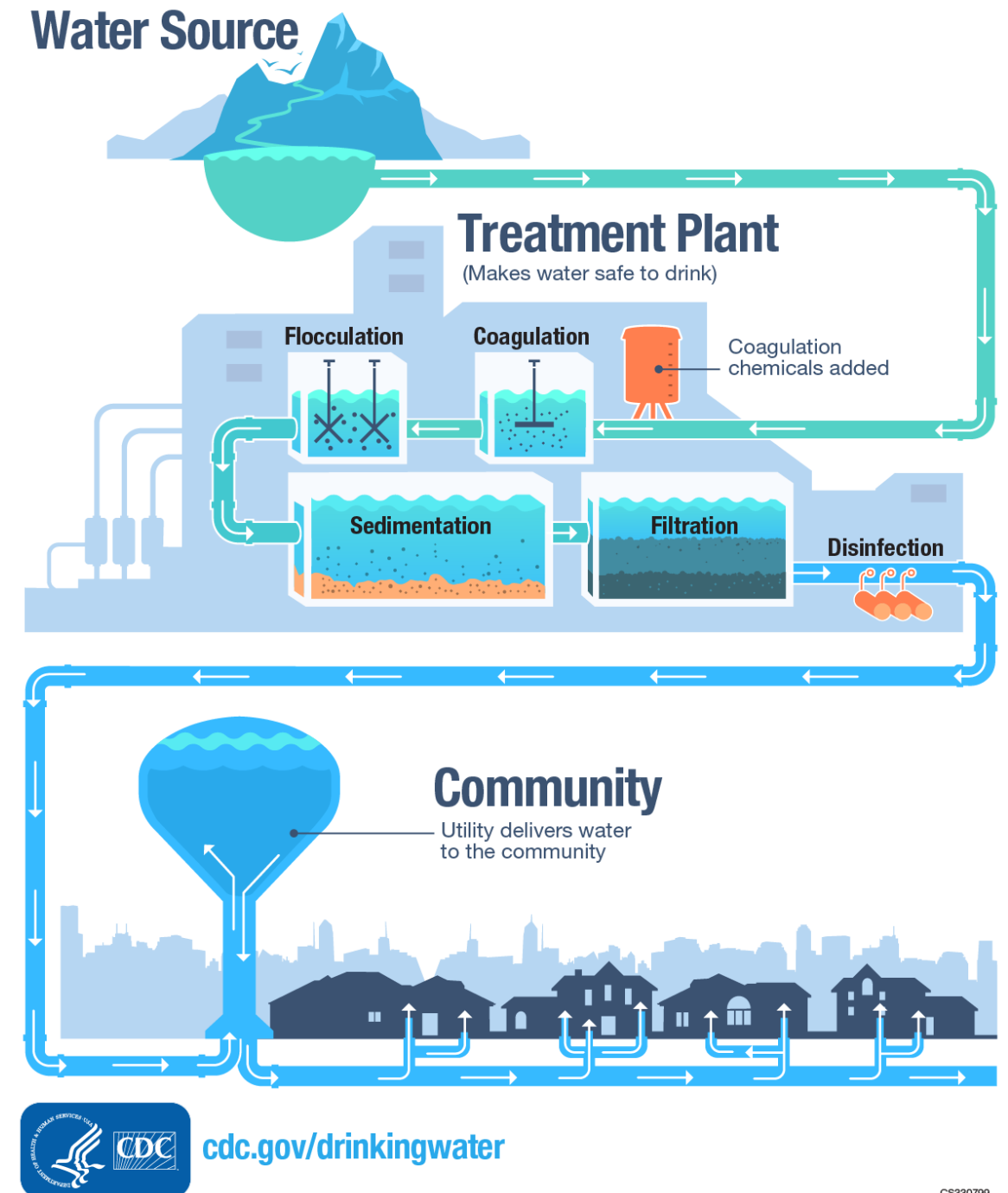
Biological: viruses, bacteria, protozoa, algae, and other small living organisms – suspended or attached

- Storage of raw water to minimize fluctuations in water quality.
- The treatment process depends on the quality of a source water.

Drinking water treatment

Common steps

- Pre-treatment/Screening
- Coagulation/Flocculation
- Sedimentation
- Filtration
- Disinfection

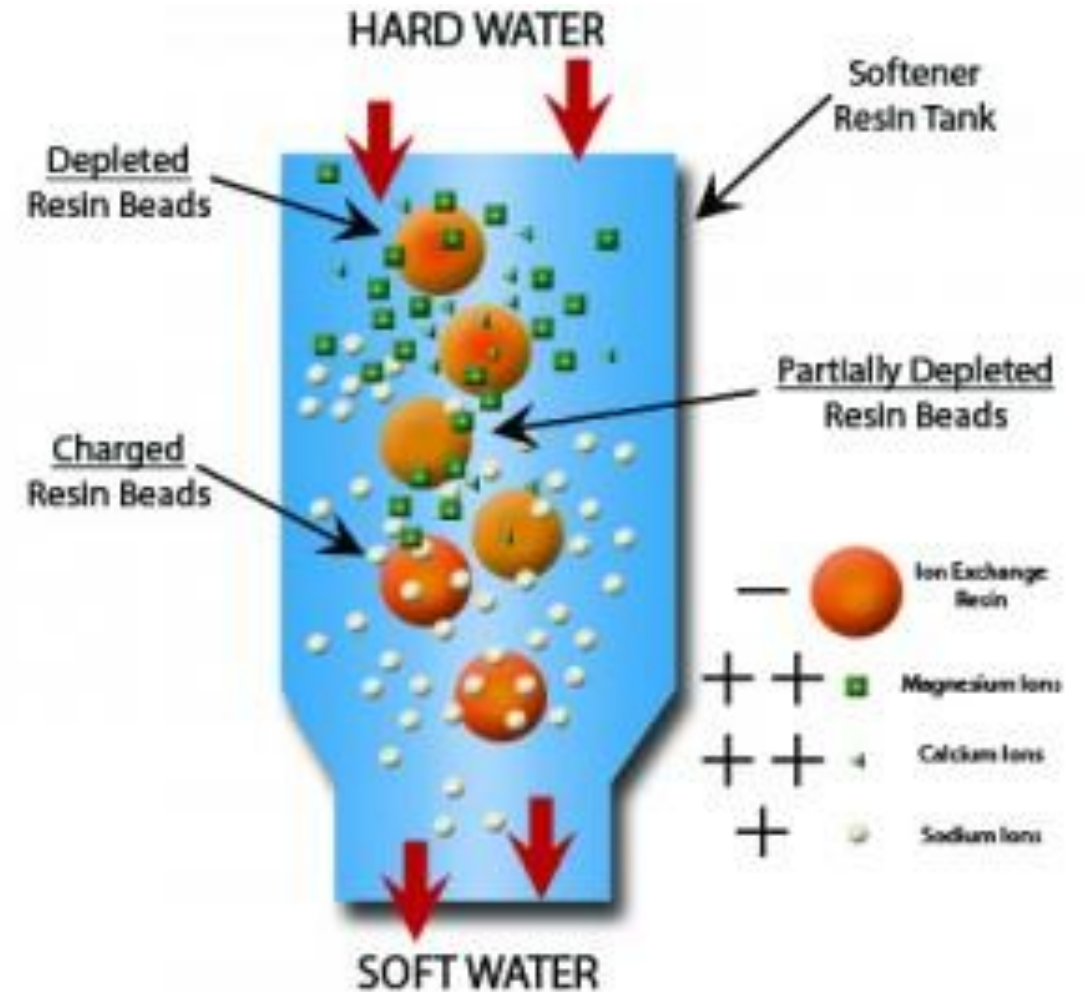


- **Sand filtration**, a simple and low-cost approach

- slow sand filtration: $0.1\text{--}1\text{ m}^3/\text{h}/\text{m}^2$
- rapid sand filtration: $5\text{--}24\text{ m}^3/\text{h}/\text{m}^2$

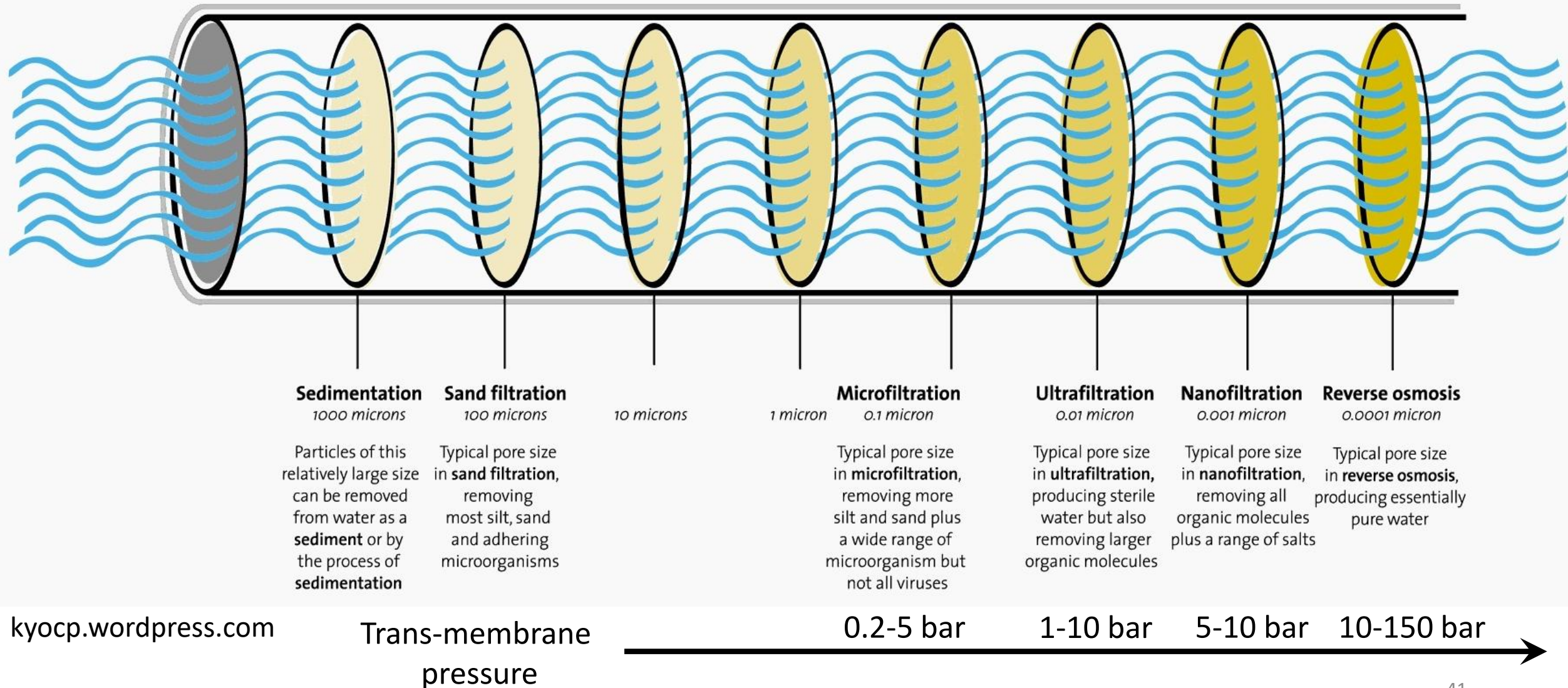
- **Ion exchange resins**

- **Membrane filtration**



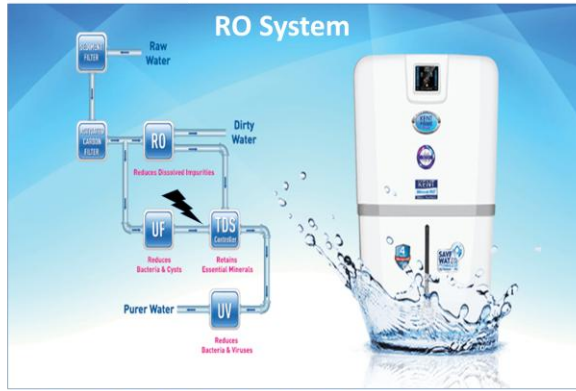
MEMBRANE FILTRATION

The pore size at each step is one-tenth smaller
1000 microns = 1 millimetre (0.001 metre)





Are we using RO wisely?



<https://www.kent.co.in/blog/purify-water-for-drinking-with-kent-reverse-osmosis-water-purifiers>

- Removes high amount of TDS
- Removes microorganisms
- Removes toxic heavy metals
- pH and taste control

When should RO system be used?

Levels of TDS mg/L	Potability	Requirement of RO system
less than 100	Not recommended	Nil
100	Acceptable	Nil
150-300	Excellent	Nil
300-600	Good	Can be used
600-900	Fair	Needed
900-1200	Poor	Needed
Above 1200	Unacceptable	Needed



RO System at IISER Mohali

- Water wastage (for 1L of drinking water 3-4 L water is wasted).
- Removes TDS below recommended level (if not regulated).
- Deficiency of iron and magnesium results in increased heart disease and affects the growth of children
- Suicidal tendencies can be increased.
- Weak immunity development in kids
(Source: Sedlak, D. ES&T, 2019)



Think of air water machine instead of RO system

Storage and usage of the RO reject water



<https://www.thebetterindia.com/92731/5-ways-to-use-ro-wastewater>



Poster made for Jal Shakti Abhiyan 2019 by Env. Microbiology lab students

Disinfection

- Essential final step to remove remaining microbes – before water is distributed for consumption
- **Key factors in selecting a disinfectant include:**
 - effectiveness in killing a wide range of microorganisms
 - ability to remain effective in the water throughout the distribution system (residual activity)
 - potential to form possibly harmful disinfection by-products – toxicity to humans and other life forms
 - safety and ease of handling chemicals and equipment
 - non-corrosive
 - cost

Disinfectants

- **Chlorination** - Chlorine, chlorine dioxide, chloramine, etc. – strong oxidizing agents, highly soluble in water, low-cost and widely used approach; residual activity in distribution network; byproducts toxic
- **Ozonation** - Ozone – stronger oxidizing properties (7X of free chlorine); no toxic byproducts, no residual activity, expensive
- **UV radiation** – Damages DNA, Effective wavelength 240 to 280 nm; no byproducts, no residual activity, affected by turbidity, and ineffective against chemical contaminants

Advanced water treatment technologies

Process	Dissolved organic matter	Trace organics	TDS*	Bacteria & protozoa	Viruses	Minerals
Microfiltration				√		
Ultrafiltration		√		√	√	
Nanofiltration	√	√	√	√	√	√
Reverse osmosis	√	√	√	√	√	√
Electrodialysis			√			
Granular activated carbon	√	√				
Ion exchange		√	√			
Advanced oxidation processes (e.g., Ozonation)		√		√	√	
Ultraviolet disinfection				√	√	

*Total dissolved solids

Water treatment at IISER Mohali

Storage & Distribution

Reservoir +
screening +
equalization



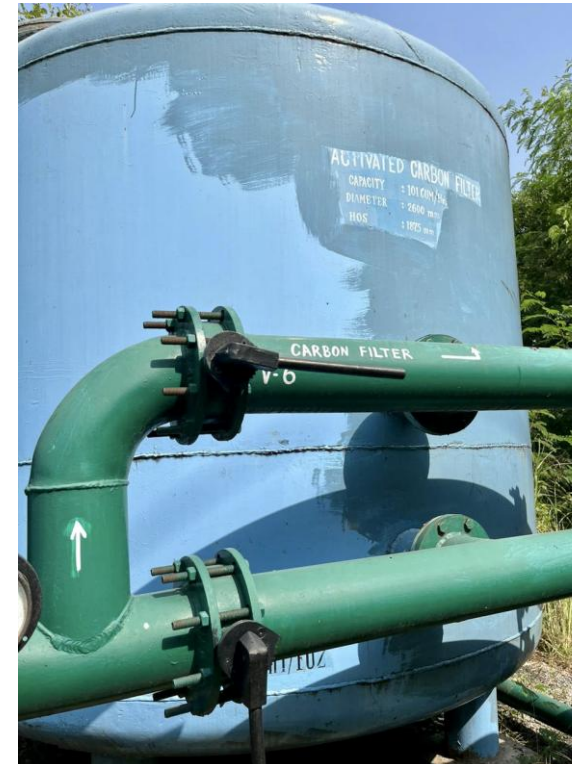
Rapid sand
filtration



Activated carbon
filtration



Ozonation



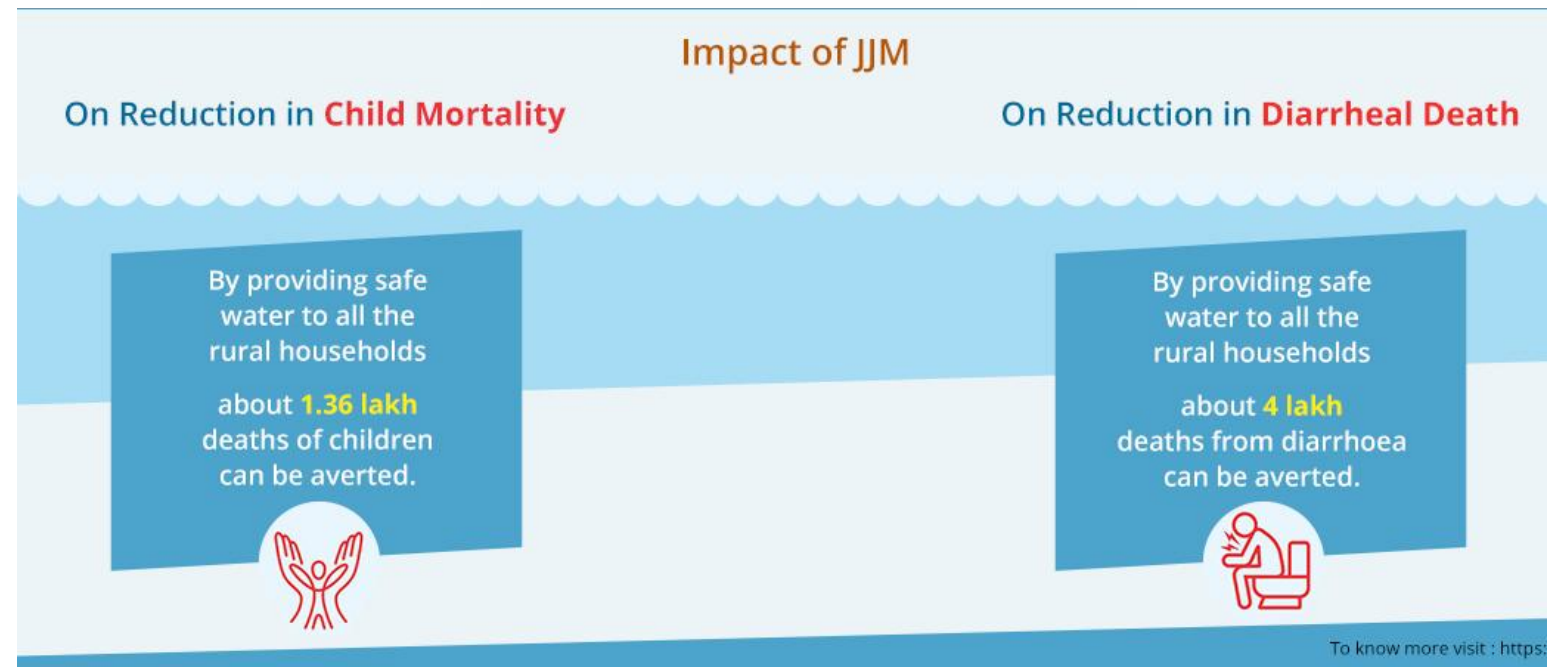
SDG 6



**Ensure availability
and sustainable
management of
water and
sanitation for all**



**Har Ghar Jal
Jal Jeevan Mission**



<https://ejalshakti.gov.in/jjmreport/JJMIndia.aspx>

Water footprint of consumer products

- **Volume of freshwater used and/or polluted to produce a product/service** - the amount of water used in various steps of the production chain of any consumer product (commodity, good, process, or service).
- Professor Arjen Hoekstra (1967 – 2019) at the University of Twente pioneered the water footprint concept in 2002 - to understand how our production and consumption choices are affecting freshwater resources.
- Useful to promote sustainable, fair, and efficient freshwater use in the production chain.

- The water footprint can be estimated for
 - well-defined group of consumers (e.g., an individual, family, village, city, province, state, or nation)
 - producers (e.g., a public organization, private enterprise, or economic sector)
 - a single process (e.g., growing rice)
 - a product's entire value chain
 - a service

Functional units: m^3 per ton of production, m^3 per hectare of cropland, m^3 per unit of currency, etc.

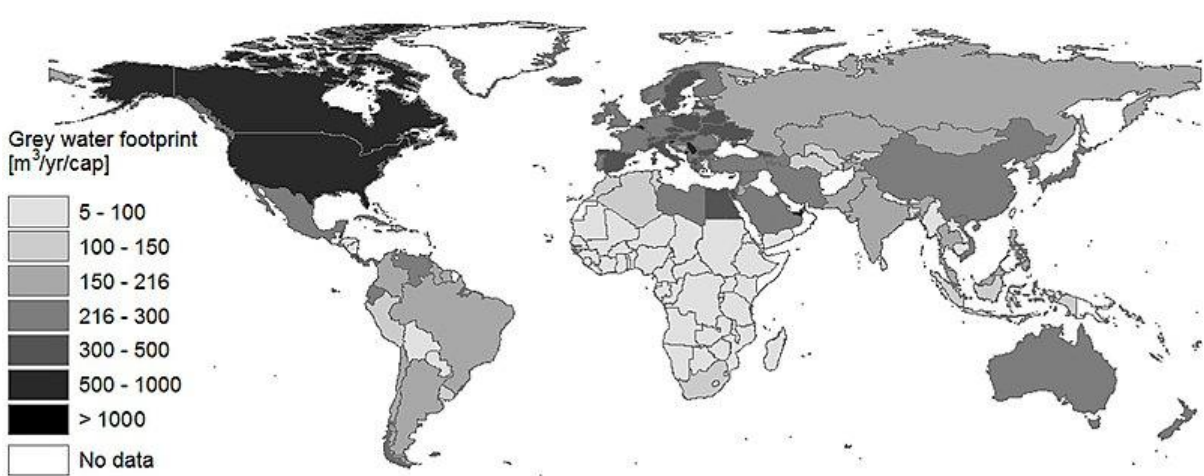
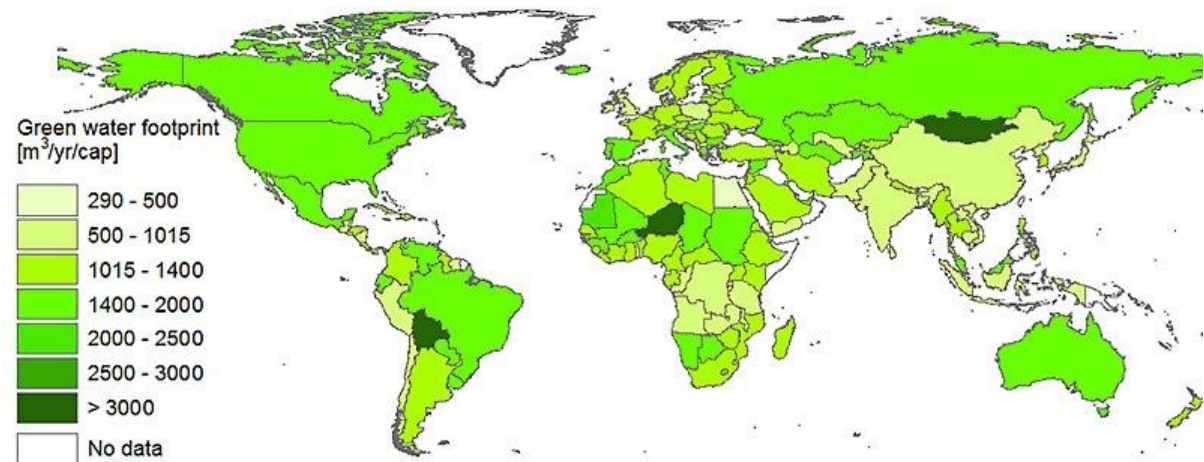
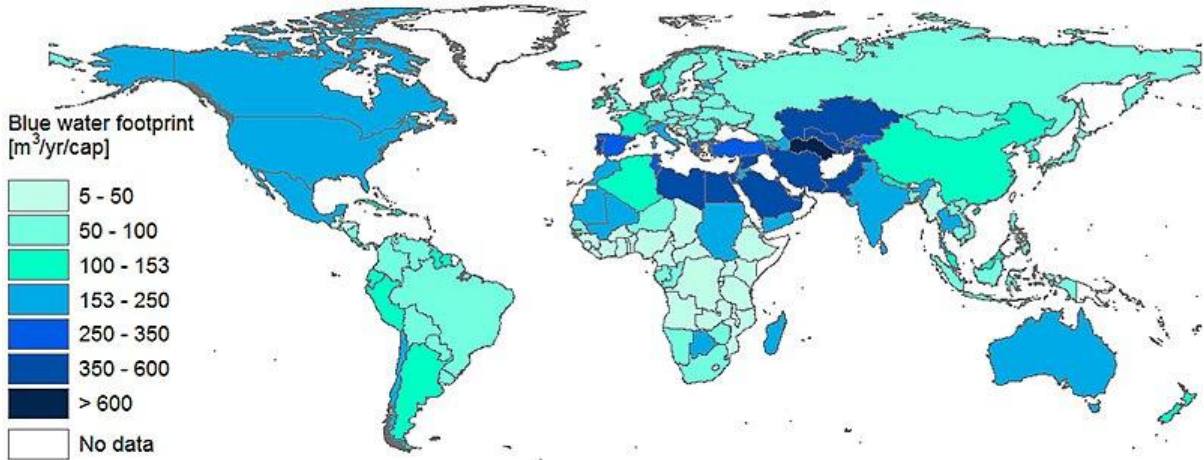
E.g., The water footprint (m^3/ton) of primary crops can be calculated:

water use at the field level (m^3/ha) / crop yield (ton/ha)

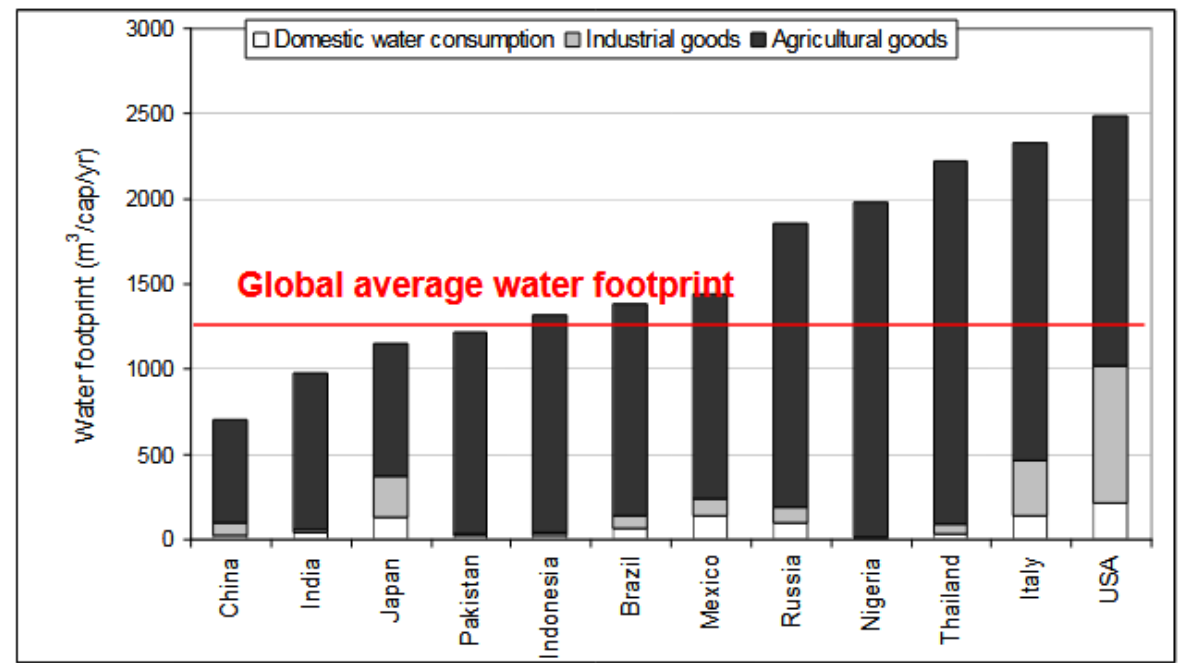
- It provides insights for
 - businesses to understand their water-related business risk
 - governments to understand the role of water in their economy and water dependency
 - consumers to know how much water is hidden in the products they use
 - **insights into whether water use is sustainable, efficient, and fair, and accordingly, ways to manage it sustainably**

Three water footprints

- **Green water footprint** – refers to rainwater
 - relevant for agricultural, horticultural, and forestry products
- **Blue water footprint** – refers to surface water or groundwater resources
 - Irrigated agriculture, industry, and domestic water use can each have a blue water footprint
- **Grey water footprint:** volume of polluted freshwater– estimated as the volume of freshwater required to dilute pollutants to meet specific water quality standards

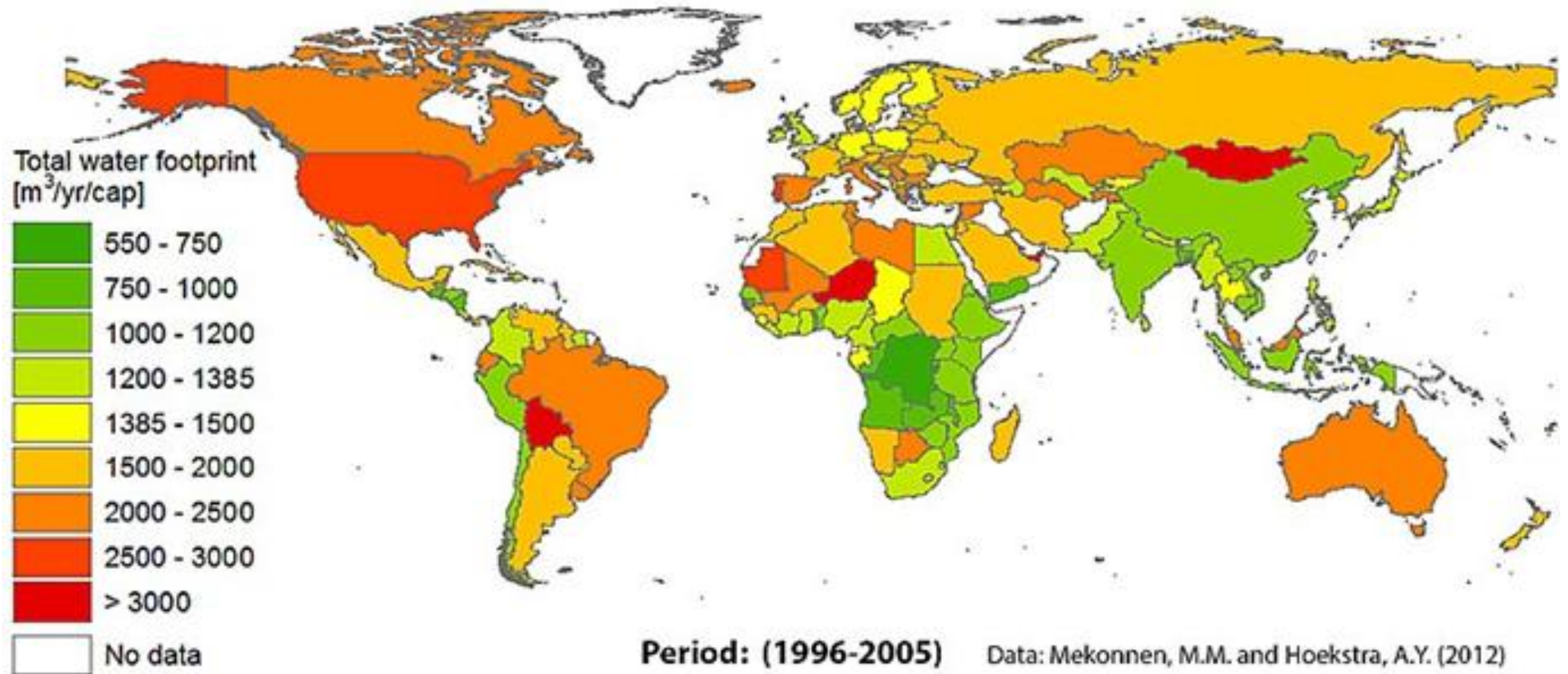


Water footprint per capita



[Hoekstra & Chapagain, 2008]

Period: (1996-2005) Data: Mekonnen, M.M. and Hoekstra, A.Y. (2012)



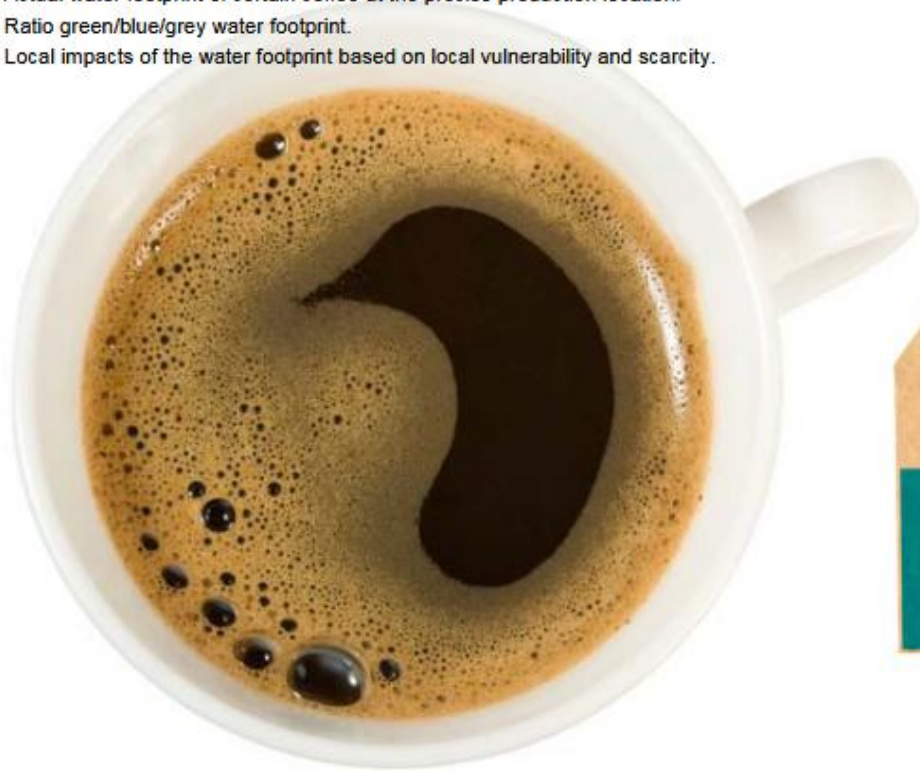
- E.g., water footprint of a pair of jeans - summation of the water footprint of each step, or process, required to produce the final product.
 - cotton to be grown
 - ginning and spinning of the fibers
 - weaving, sewing, and wet processing of the fabric
 - each step has direct and indirect water footprints

About 10,000 Liters per pair of jeans, considering all steps.

- The average individual's water footprint is approximately 3,400 liters per day.

This is a **global average** and **aggregate** number. Policy decisions should be taken on the basis of:

1. Actual water footprint of certain coffee at the precise production location.
2. Ratio green/blue/grey water footprint.
3. Local impacts of the water footprint based on local vulnerability and scarcity.



[Hoekstra & Chapagain, 2008]



[Hoekstra & Chapagain, 2008]

WATER FOOTPRINT

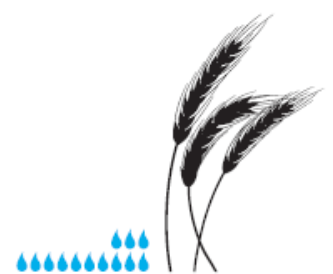
Virtual water embedded in products

→ For the full poster featuring many more products and in-depth information, visit: www.virtualwater.eu

ONE DROP (shown in the illustrations) is equivalent to 50 litres of virtual water (production-site definition). All figures shown on this poster are based on **exemplary calculations** and may vary depending on the origin and production process of the product.

The **water footprint** of a product (a commodity, good or service) is the **volume of freshwater used to produce the product**, measured at the place where the product was actually made. It refers to the amount of the water used in the various steps of the production chain.

DATA: Hoekstra, A.Y.; Chapagain, A.K. (2008) *Globalization of water: Sharing the planet's freshwater resources* Blackwell Publishing, Oxford, UK www.waterfootprint.org
DESIGN: Timm Kekeritz, www.virtualwater.eu
TYPEFACE: TheSans and TheSerif, Luc(as) de Groot



650 Barley
litres of water for one pound (500g)



650 Wheat
litres of water for one pound (500g)



1400 Sorghum
litres of water for one pound (500g)



2500 Millet
litres of water for one pound (500g)



650 Toast
litres of water for one package (500g)



750 Cane Sugar
litres of water for one package (500g)



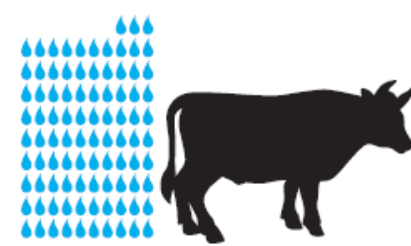
90 Tea
litres of water for one pot (750ml)



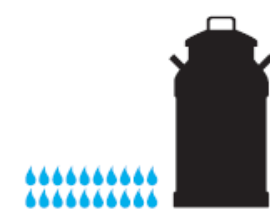
840 Coffee
litres of water for one pot (750ml)



2500 Burger
litres of water for one burger (150g beef)



4650 Beef
litres of water for one steak (300g)



1000 Milk
litres of water for one litre



2500 Cheese
litres of water for one big piece (500g)

Major determinants of water footprint

- Consumption characteristics
 - Consumption volume
 - Consumption pattern
- Production circumstances
 - Climate: evaporative demand at the place of production
 - Agricultural practice: water use efficiency
- **Efforts to reduce consumption (direct/indirect) and achieve sustainable, equitable, and efficient use of water in all sectors.**

For more information, you can refer to

- <https://waterfootprint.org/en/>
- <http://virtualwater.eu/>
- <https://youtu.be/b1f-G6v3voA> (A nice short film by UNESCO WWAP and produced and animated by Steve Cutts on water sources, use, footprint, etc.)
- <https://www.waterfootprint.org/resources/interactive-tools/extended-water-footprint-calculator/>